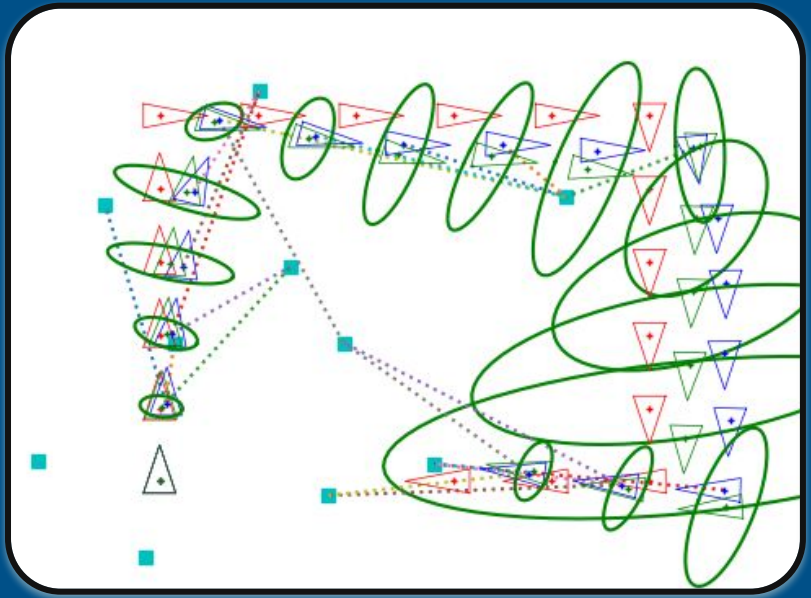
I402: Principios de la Robótica Autónoma

Clases Prácticas
Tutorial 7



Universidad de SanAndrés

Recap

Filtro de Bayes

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Recap

Filtro de Bayes

$$Bel(x_t) = \eta P(z_t \mid x_t) \underbrace{\int P(x_t \mid u_t, x_{t-1})}_{\text{Modelo de Odometría}} Bel(x_{t-1}) dx_{t-1}$$

Recap

Filtro de Bayes

$$Bel(x_t) = \eta \underline{P(z_t \mid x_t)} \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Pesos/ponderación/weights

Recap

Filtro de Bayes

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

(Normalización)

Recap

Filtro de Bayes

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

(Normalización)

Modelo de Odometría

Pesos/ponderación/weights

Recap

Modelos de Mediciones

Algorithm likelihood_field_range_finder_model(z_t, x_t, m)**:**

$q = 1$

for all k *do*

  if $z_t^k \neq z_{\max}$

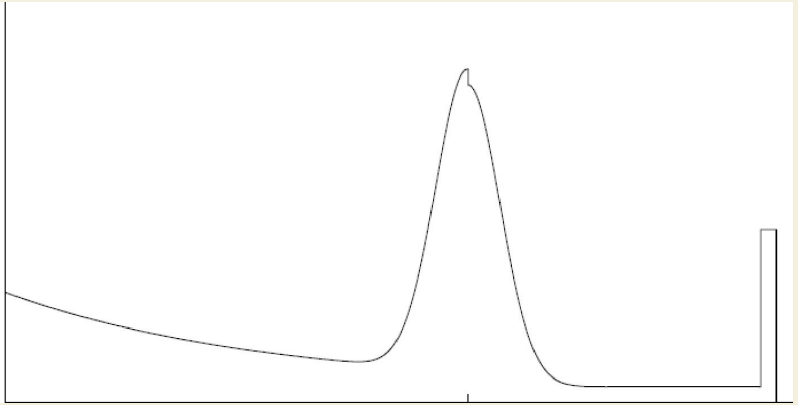
    $x_{z_t^k} = x + x_{k,\text{sens}} \cos\theta - y_{k,\text{sens}} \sin\theta + z_t^k \cos(\theta + \theta_{k,\text{sens}})$

    $y_{z_t^k} = y + y_{k,\text{sens}} \cos\theta + x_{k,\text{sens}} \sin\theta + z_t^k \sin(\theta + \theta_{k,\text{sens}})$

    $dist = \min_{x',y'} \left\{ \sqrt{(x_{z_t^k} - x')^2 + (y_{z_t^k} - y')^2} \,\middle|\, \langle x', y' \rangle \text{ occupied in } m \right\}$

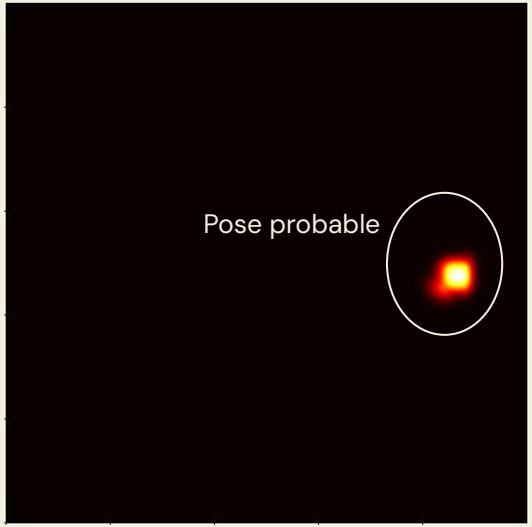
    $q = q \cdot \left(z_{\text{hit}} \cdot \mathbf{prob}(dist, \sigma_{\text{hit}}) + \frac{z_{\text{random}}}{z_{\max}} \right)$

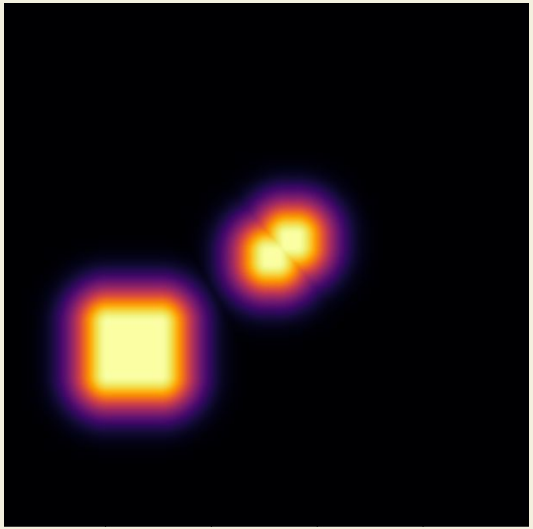
return q

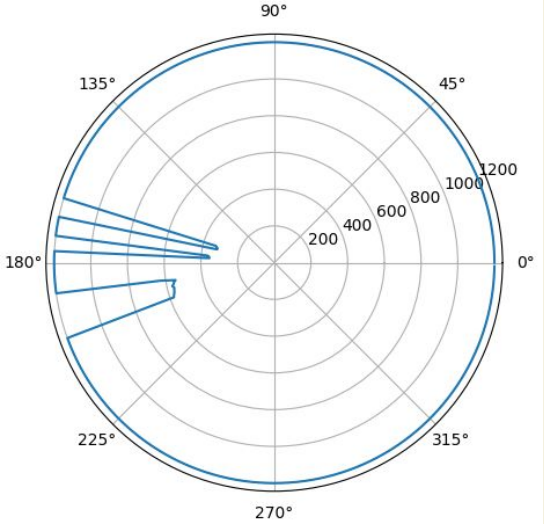


Recap

Likelihood Field







Features

Marcadores (o landmarks)

- Balizas activas (radio, GPS)
- Marcadores Pasivos (visual, retro-reflexivo)
- Método más común: triangulación

- El sensor provee
  - distancia
  - orientación
  - distancia y orientación.

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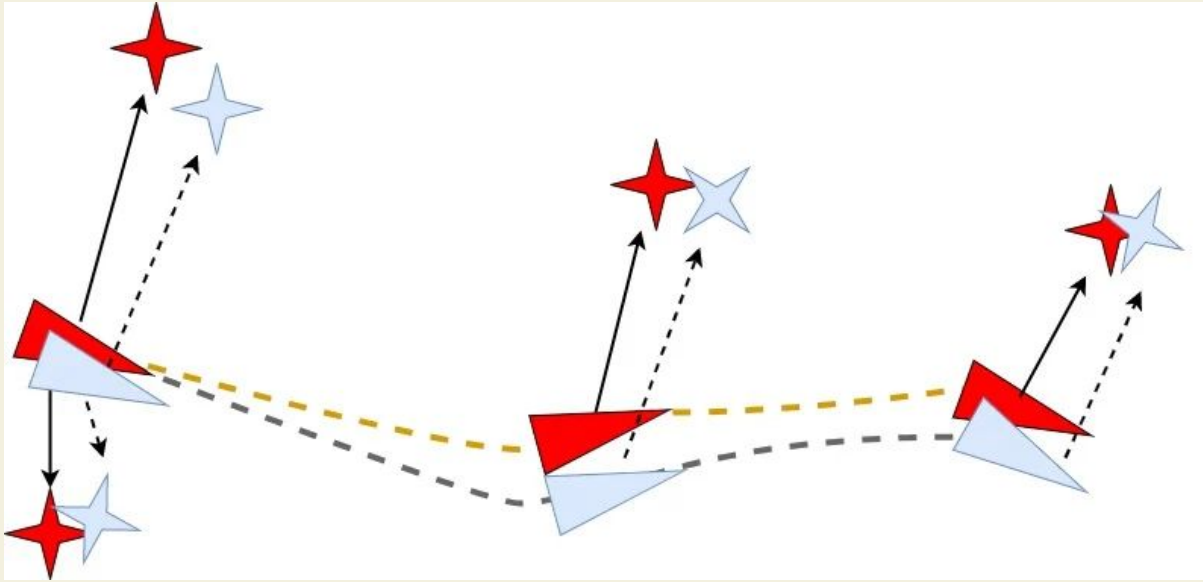
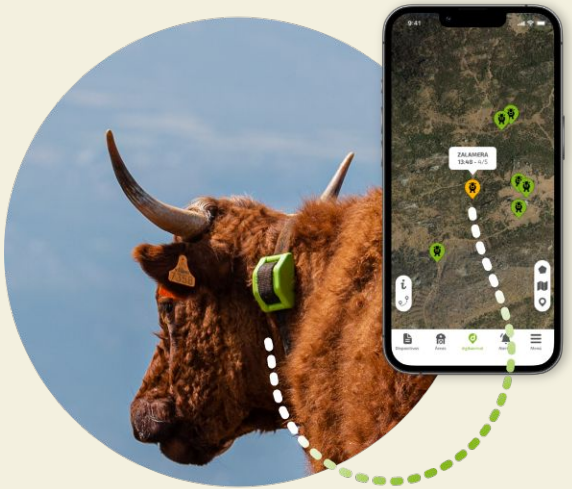
Features



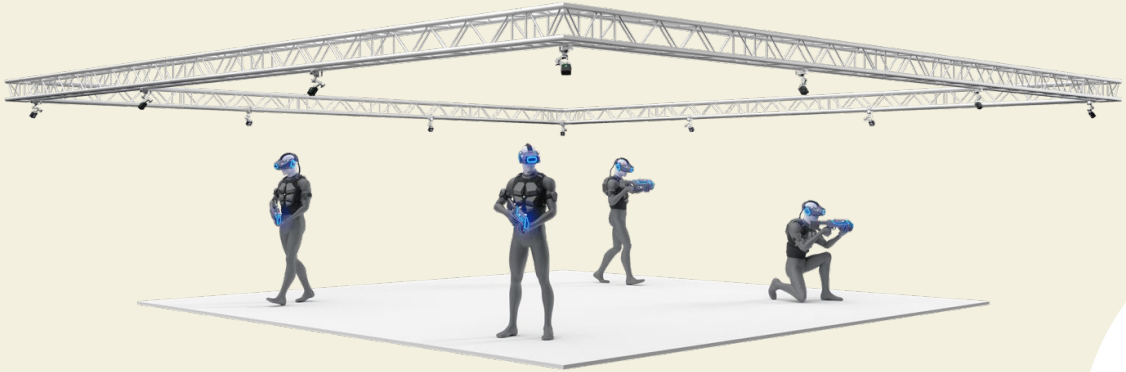
Features - Origen

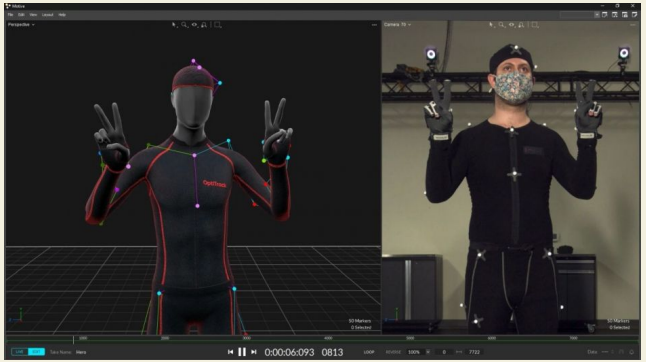
- GPS/Radio - Balizas activas



Features - Origen

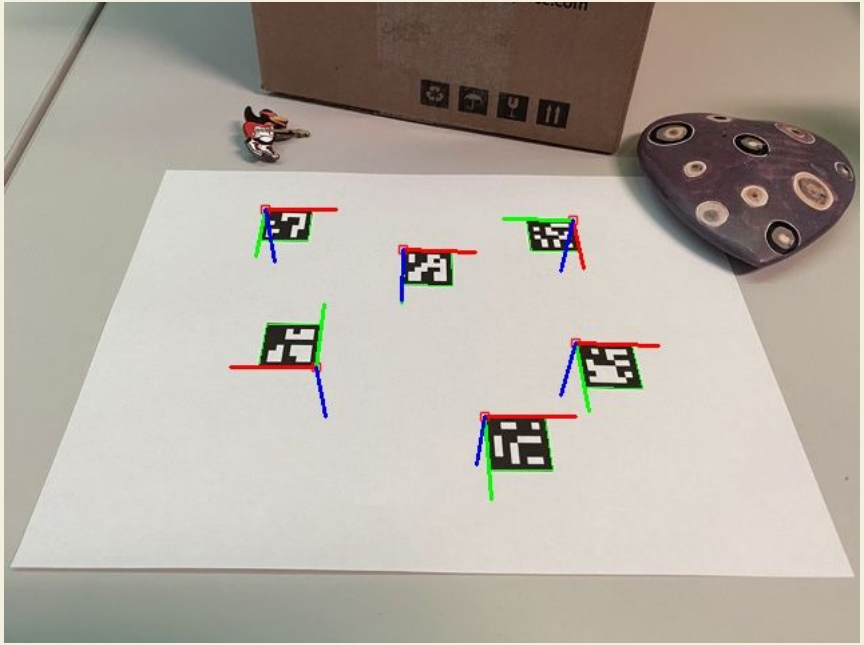
- GPS/Radio - Balizas activas
- Infrarrojo - Marcadores pasivos retro-reflexivos





Features - Origen

- GPS/Radio - Balizas activas
- Infrarrojo - Marcadores pasivos retro-reflexivos
- Visual - Tags

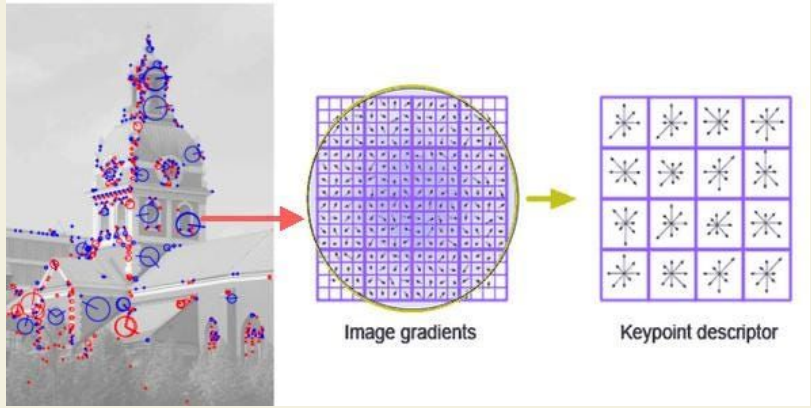


Features - Origen

- GPS/Radio - Balizas activas
- Infrarrojo - Marcadores pasivos retro-reflexivos
- Visual - Tags
- Extracción de Features

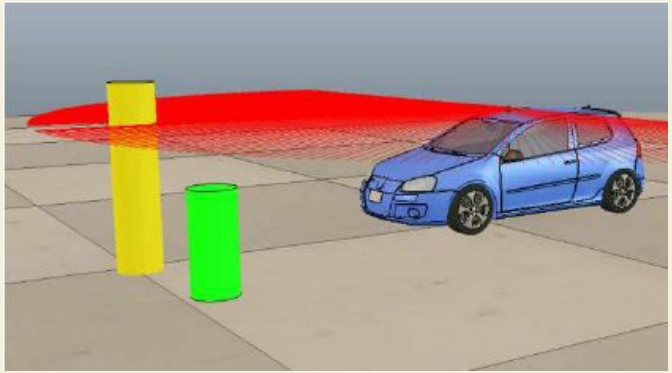
Features - Origen

- GPS/Radio - Balizas activas
- Infrarrojo - Marcadores pasivos retro-reflexivos
- Visual - Tags
- Extracción de Features
  - Visual

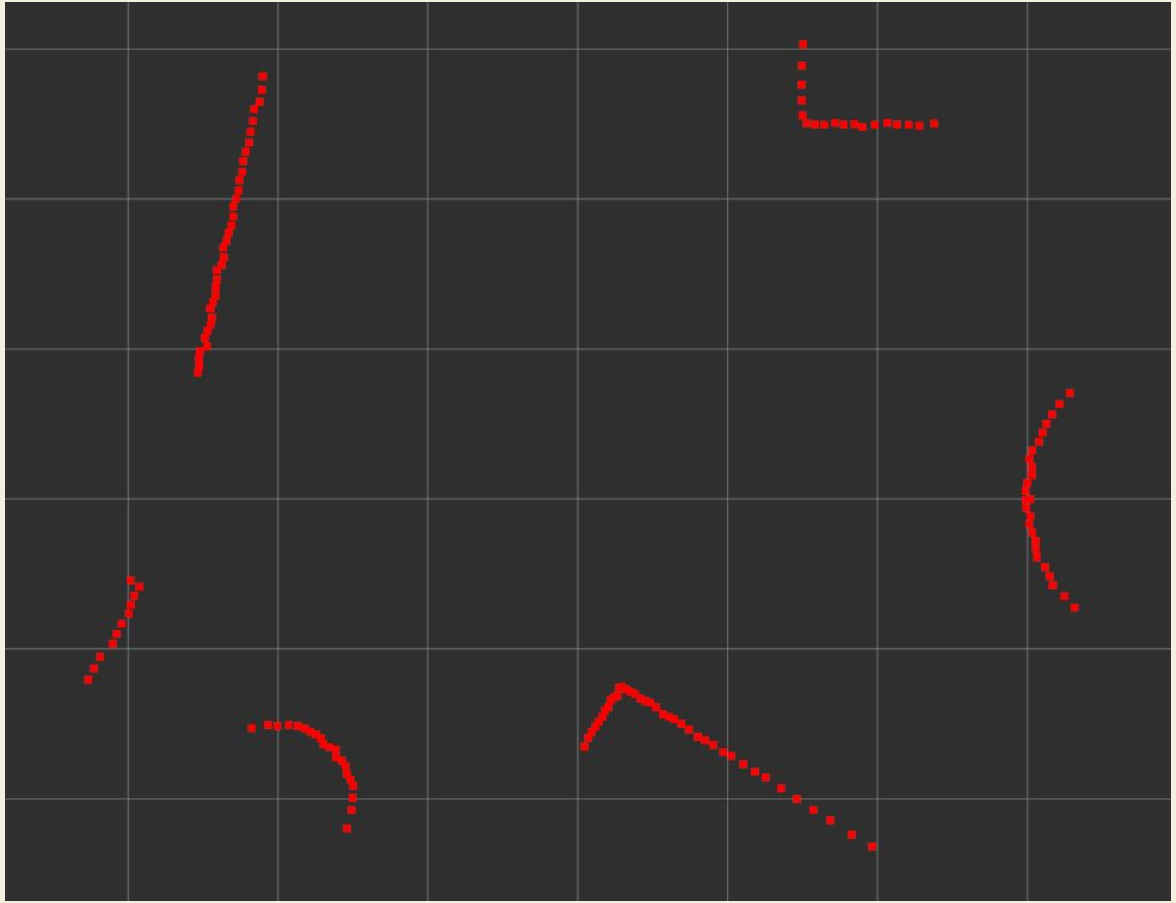


Features - Origen

- GPS/Radio - Balizas activas
- Infrarrojo - Marcadores pasivos retro-reflexivos
- Visual - Tags
- Extracción de Features
  - Visual
  - LIDAR/Ultrasonido



Features - Extracción



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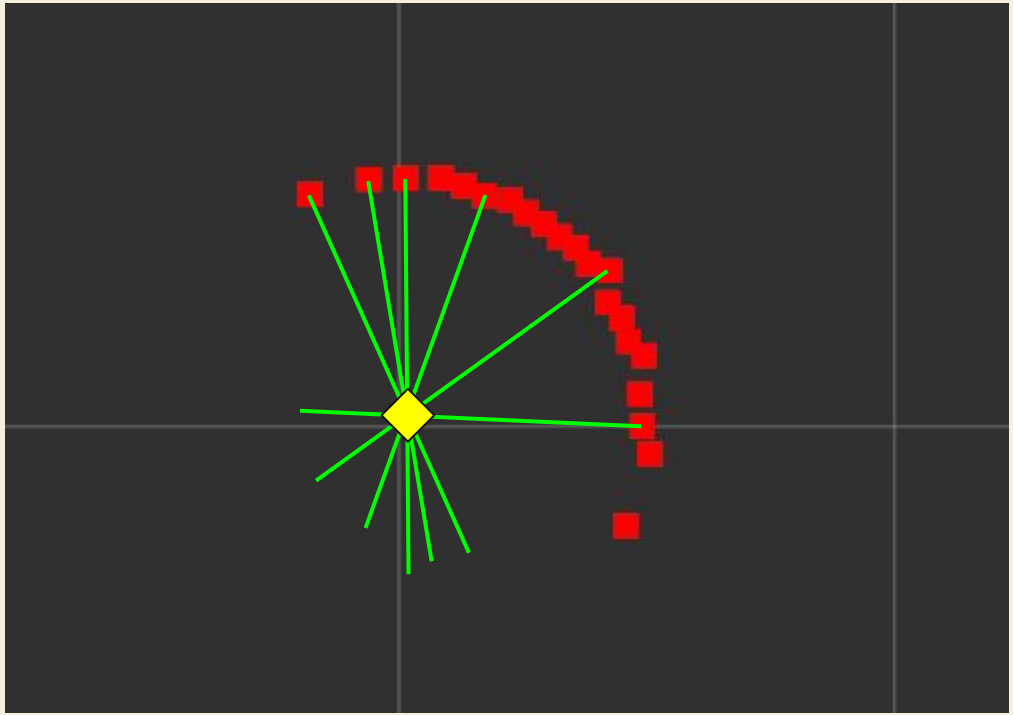

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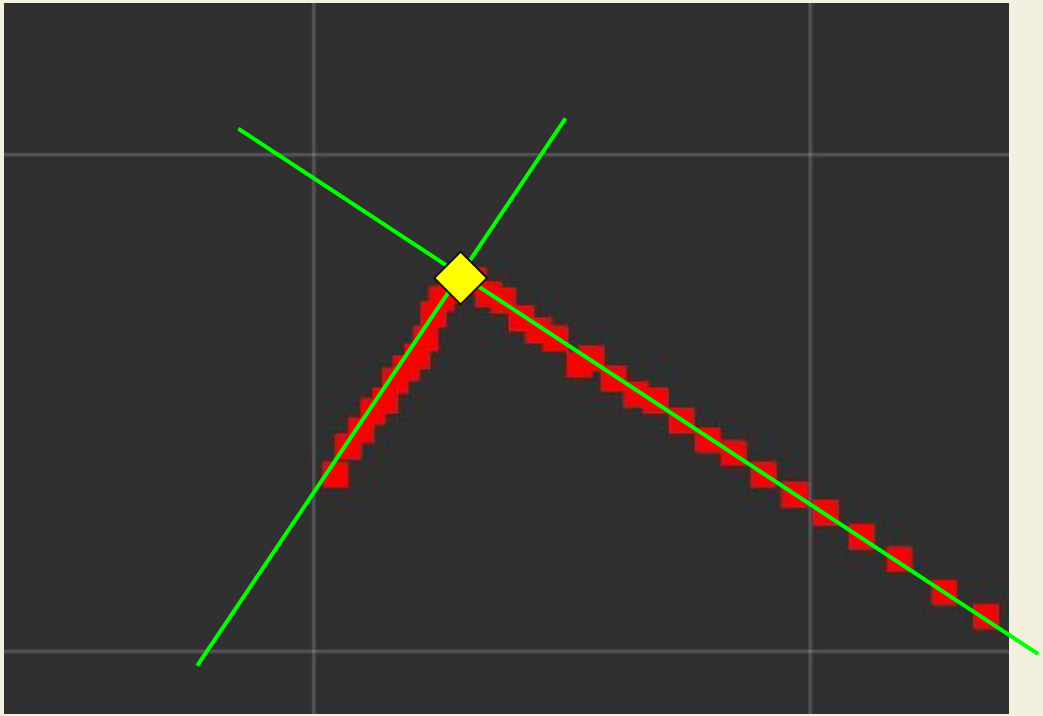

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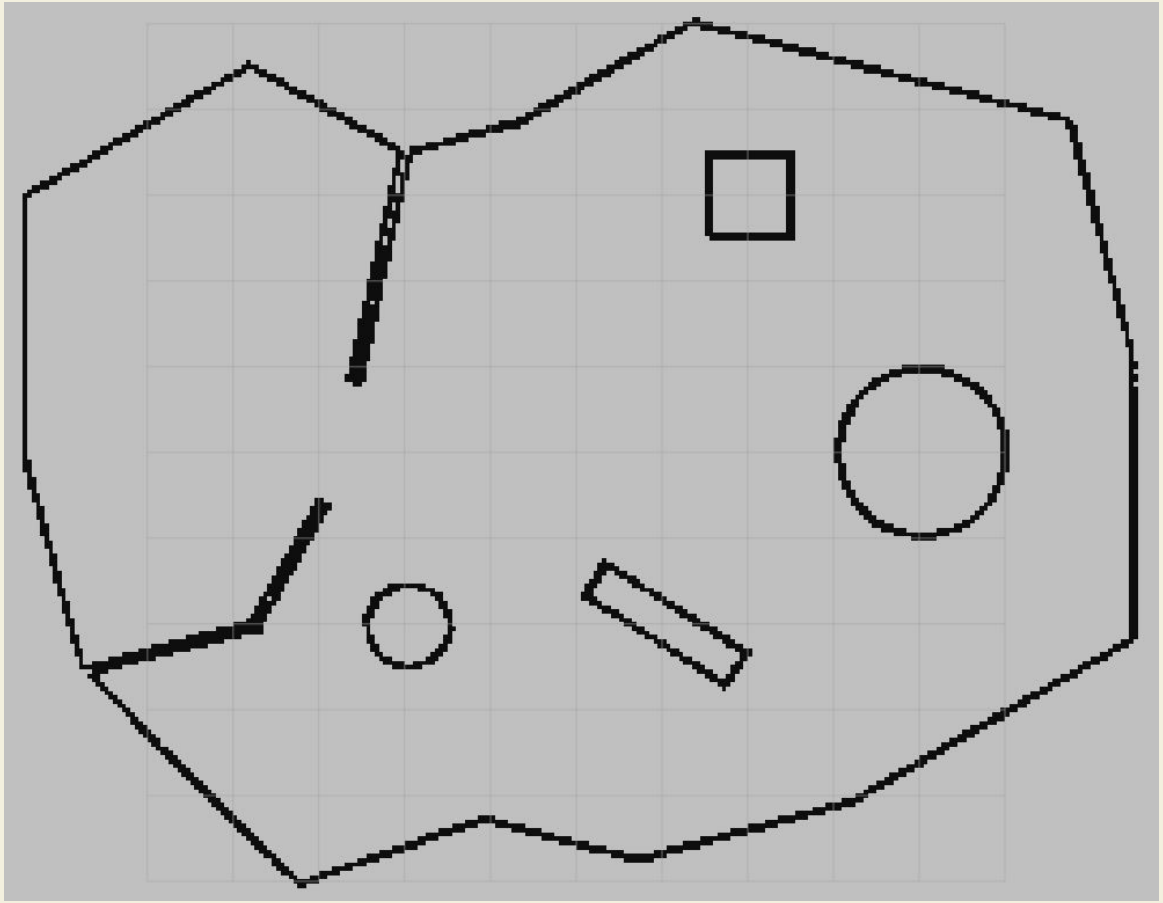

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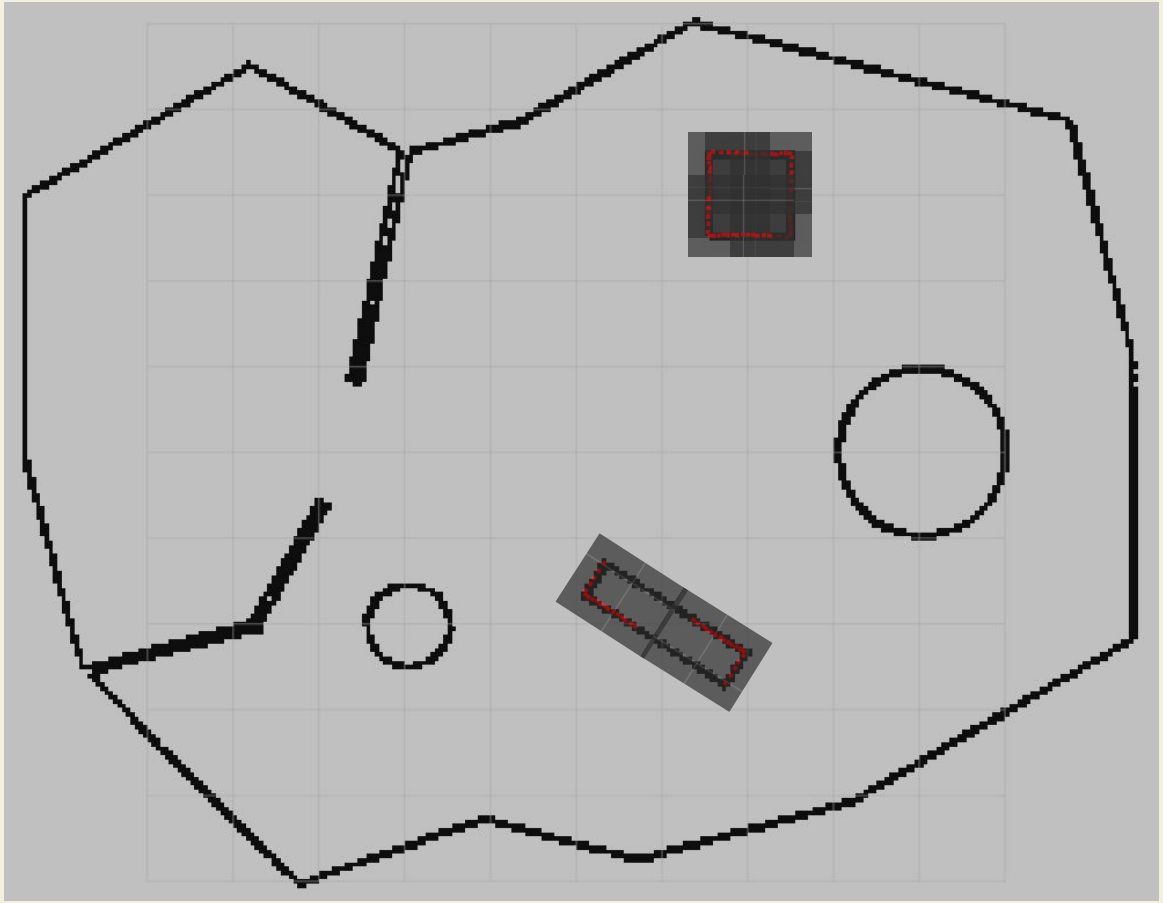
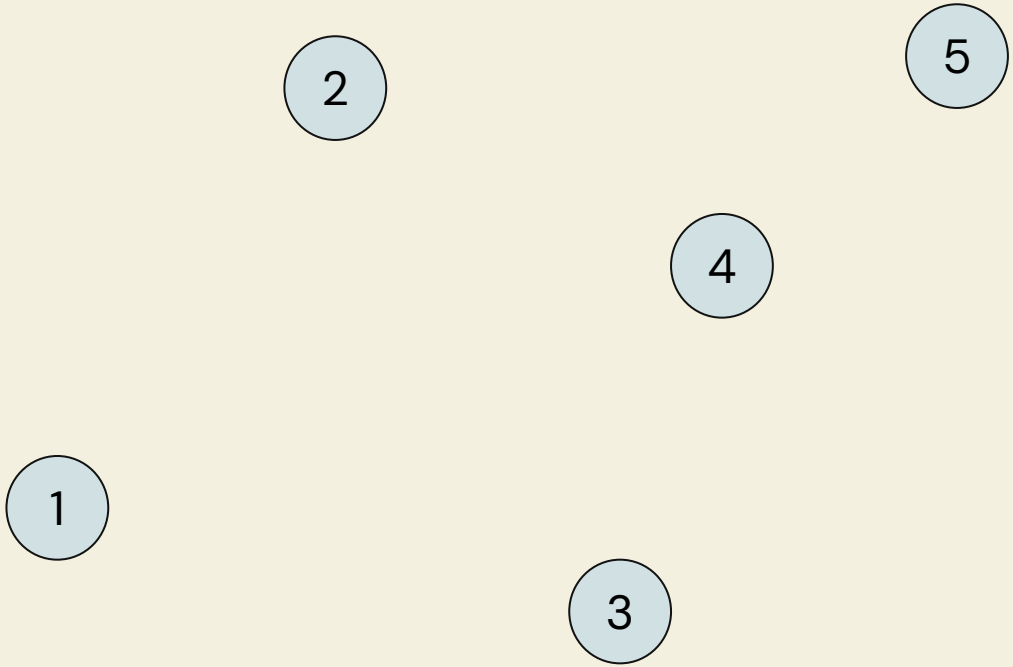


Features – Nearest Neighbour



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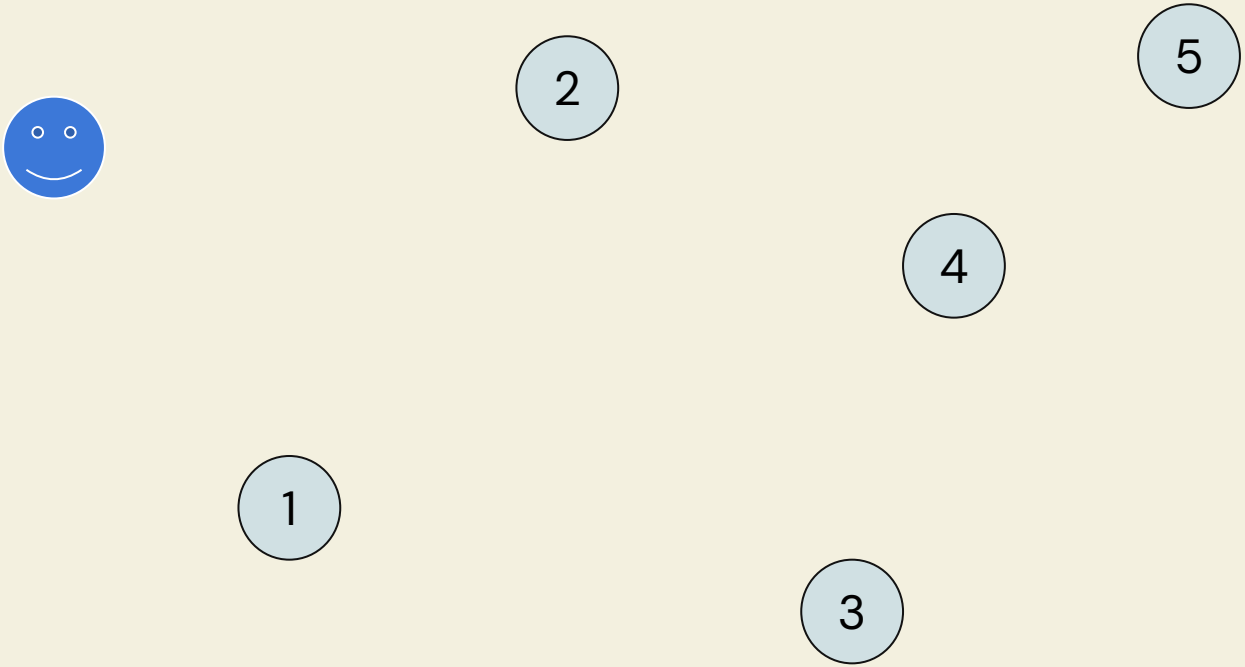

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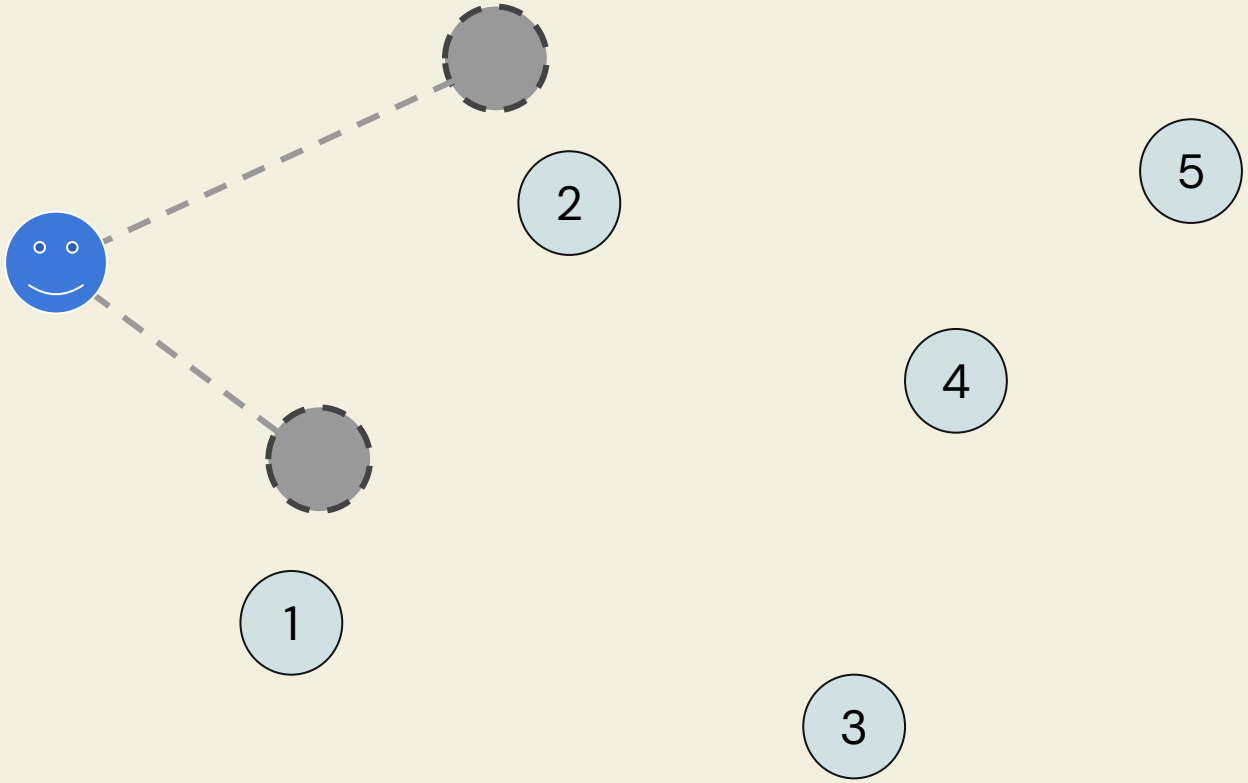

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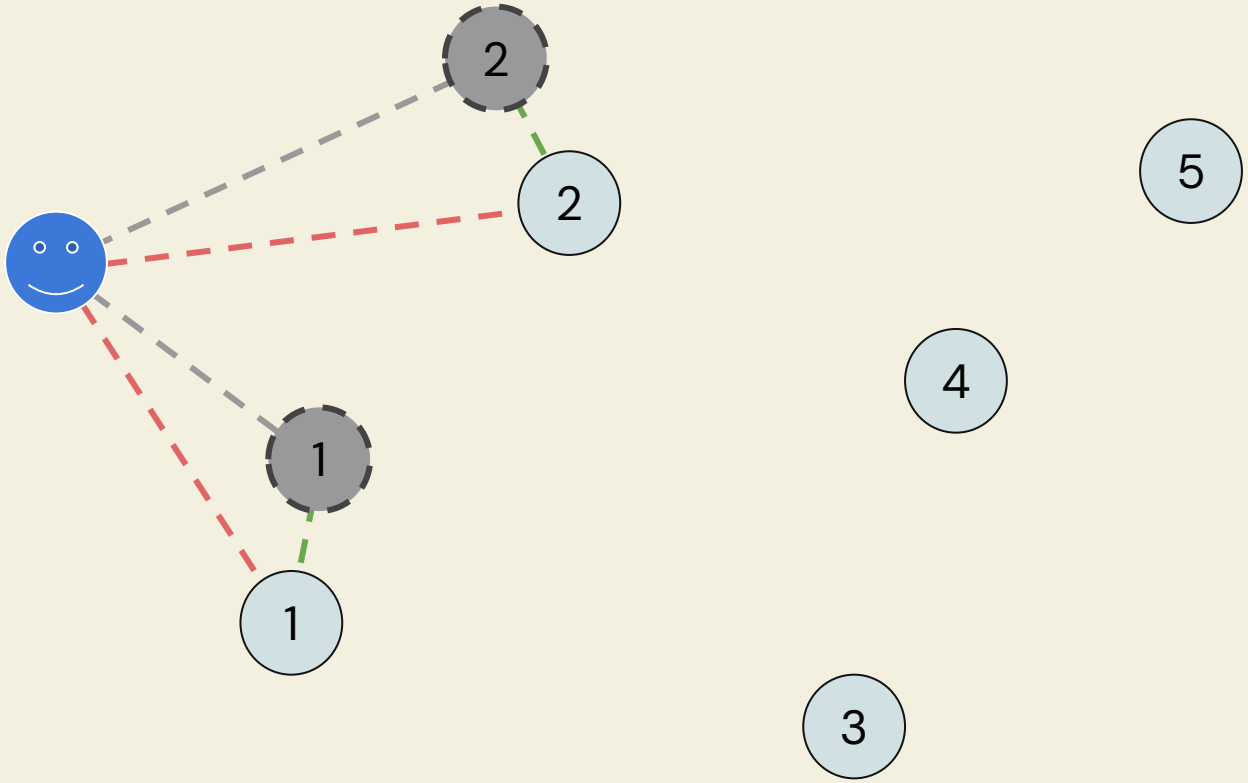
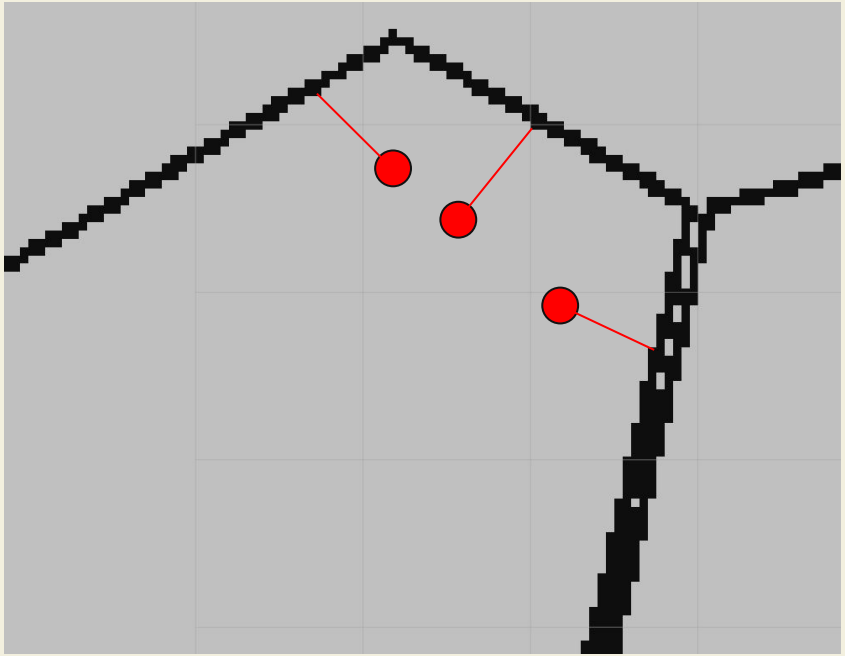
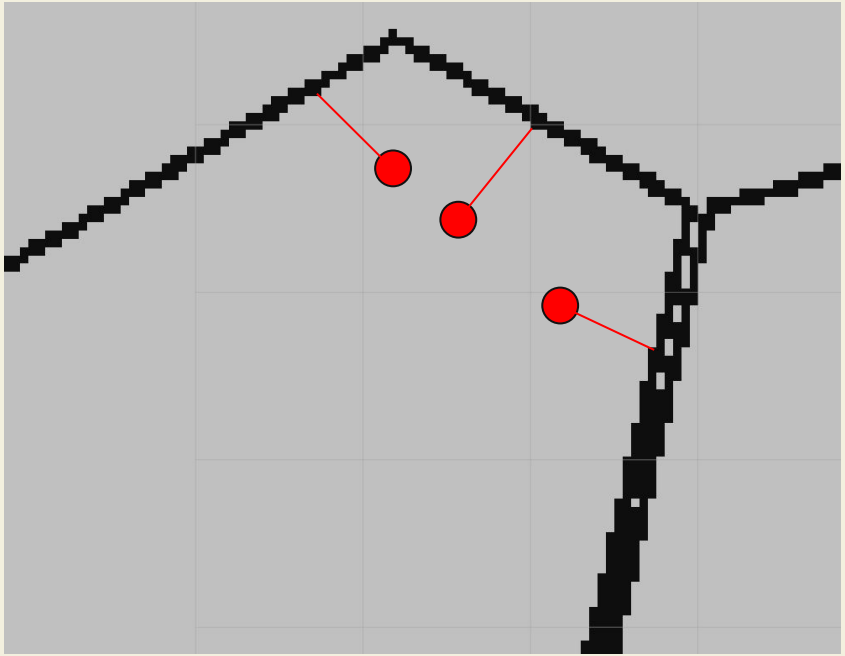


Features – LIDAR

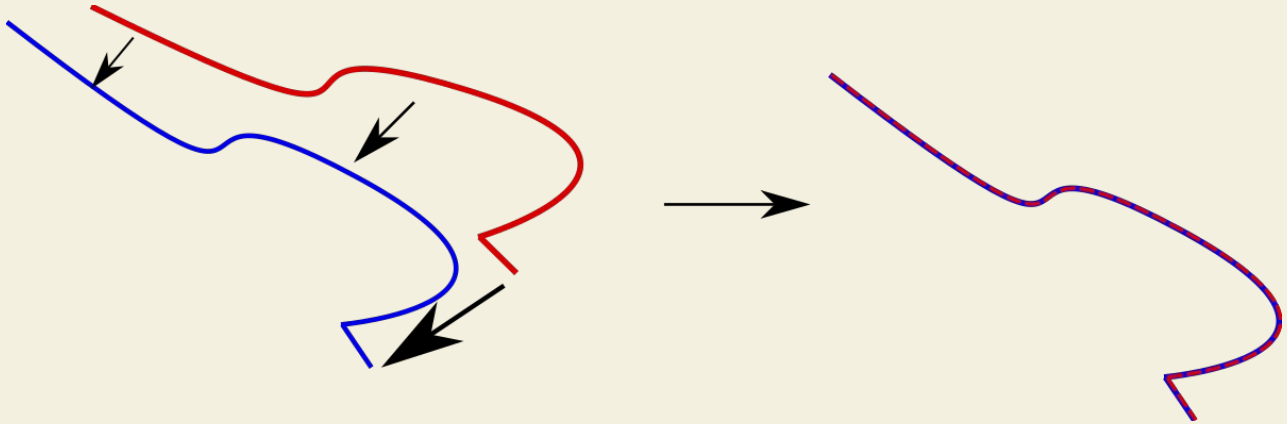


Features - LIDAR



Mejor si extraigo features del lidar, pero me tengo que asegurar que extraigo las mismas features del mapa

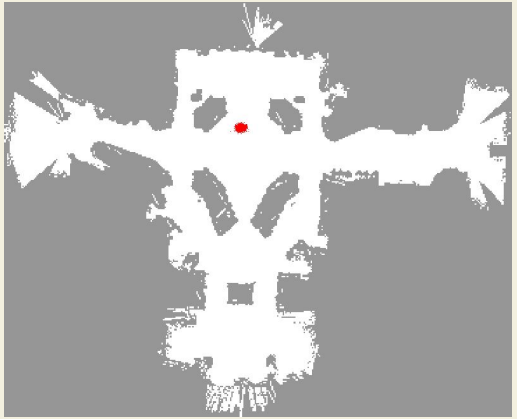
Features – ICP

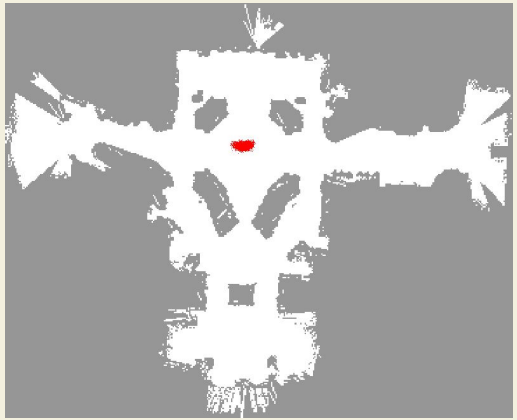


Hoy

Filtro de Partículas

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):
- $\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$
- *for* $m = 1$ *to* M *do*
  - *sample* $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$
  - $w_t^{[m]} = p(z_t \mid x_t^{[m]})$
  - $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$
- *endfor*
- *for* $m = 1$ *to* M *do*
  - *draw* i *with probability* $\propto w_t^{[i]}$
  - *add* $x_t^{[i]}$ *to* $\mathcal{X}_t$
- *endfor*
- *return* $\mathcal{X}_t$





Filtro de Partículas (TP3)

- **“Filtro”**: es un algoritmo que procesa una lectura cruda de **sensado** y **actuado**, y según su información pasada devuelve una información precisa/filtrada.
- **“Filtro no paramétrico”** → Su definición no exige una distribución (Kalman sí → paramétrico)
- “**Partículas”**: óptimo para simular estocasticidad sin que su requisito de cómputo escale con la complejidad/multidimensionalidad del problema.
- **“Resource-adaptive algorithm”** → Si tengo pocos recursos en cierto t, el algoritmo puede bajar el número de partículas a cambio de menor precisión.
- Las partículas agregan cierto error **saludable** al problema.
- El resampleo asegura buena **calidad** de partículas.

Filtro de Partículas: Predicción

Algorithm Particle_filter$(\mathcal{X}_{t-1}, u_t, z_t)$**:**

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endfor

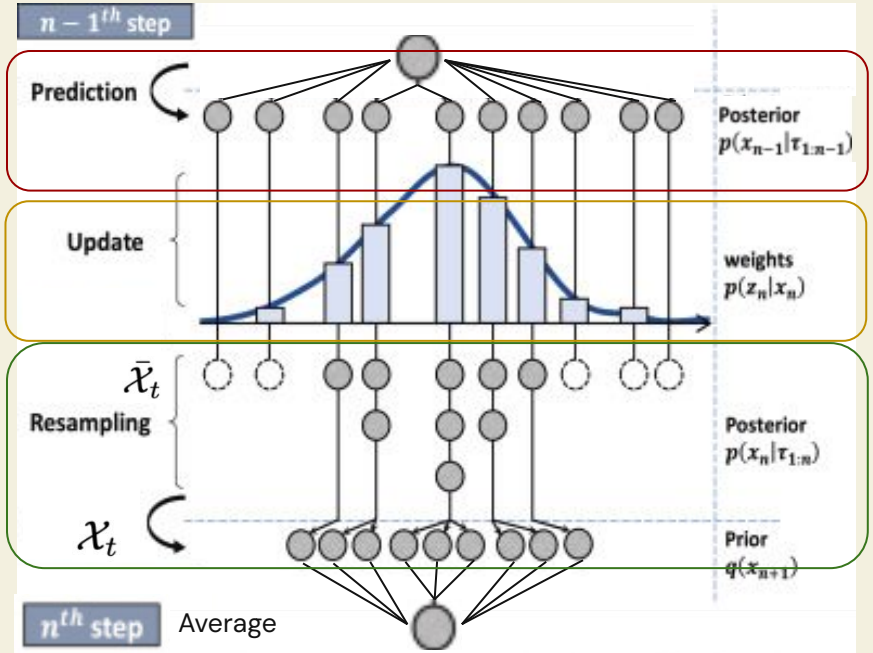
for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$



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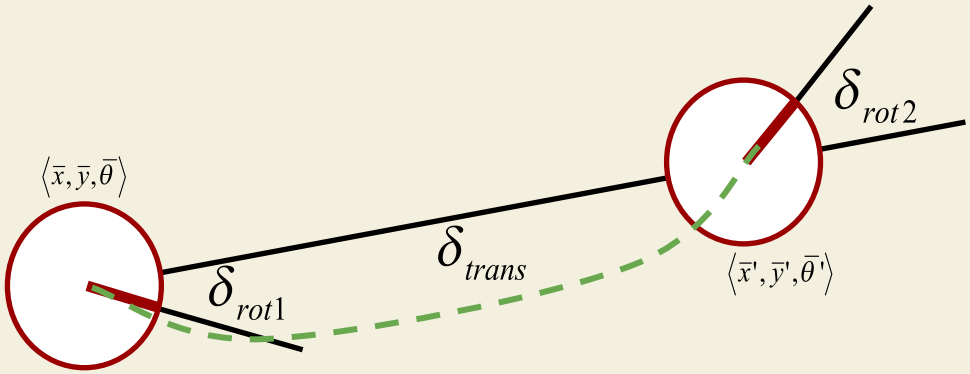
Modelos de Odometría



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endfor

for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

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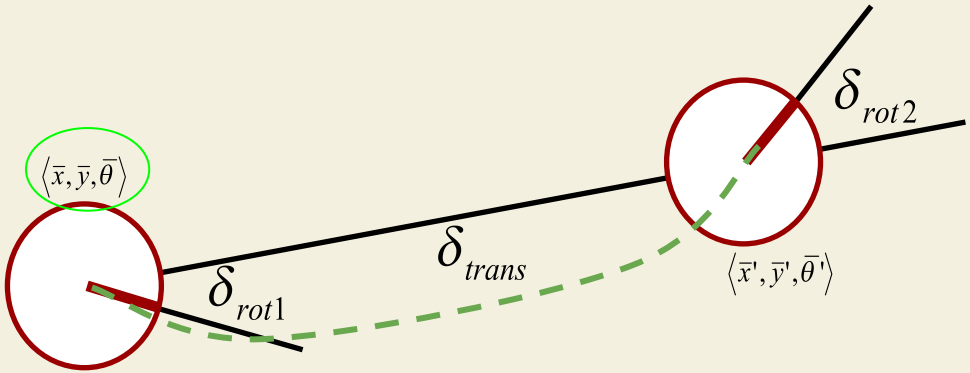
endfor

return $\mathcal{X}_t$

Modelos de Odometría



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Filtro de Partículas: Predicción

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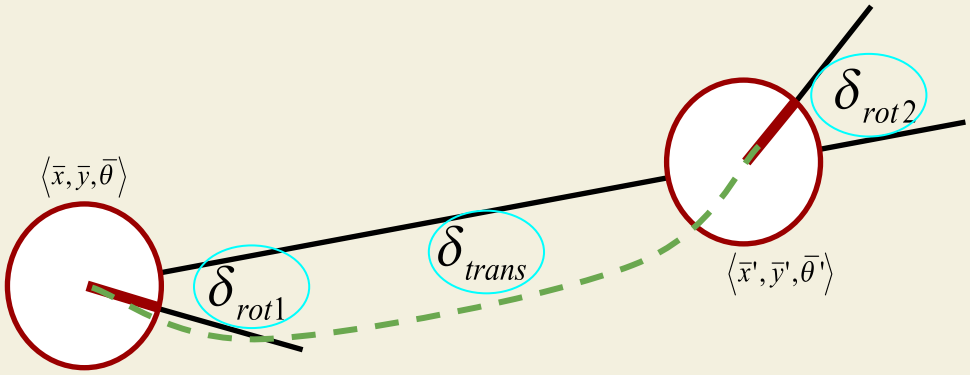
Modelos de Odometría



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Filtro de Partículas: Predicción

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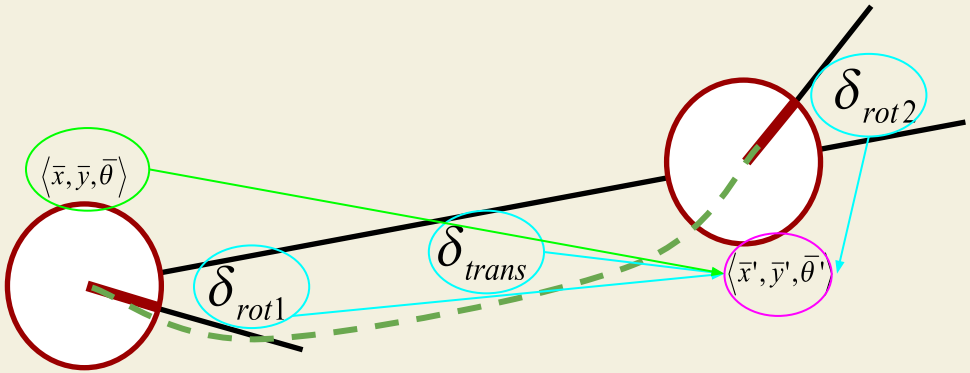
Modelos de Odometría



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Filtro de Partículas: Predicción

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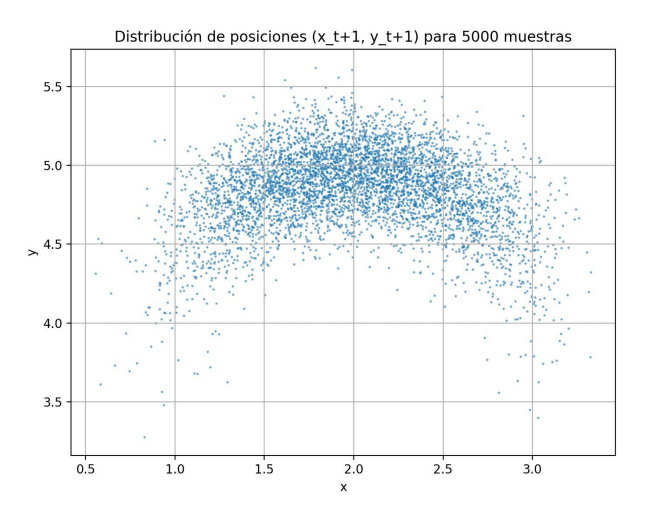
Modelos de Odometría (TP2)



$$Bel(x_t) = \eta P(z_t \mid x_t) \boxed{\int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}}$$

Filtro de Partículas: Actualización

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

$\boxed{w_t^{[m]} = p(z_t \mid x_t^{[m]})}$

$\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

draw i *with probability* $\propto w_t^{[i]}$

add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

La probabilidad de **medición** dada la **partícula** → Elegir distribución.

$$Bel(x_t) = \eta \boxed{P(z_t \mid x_t)} \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Actualización

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endfor

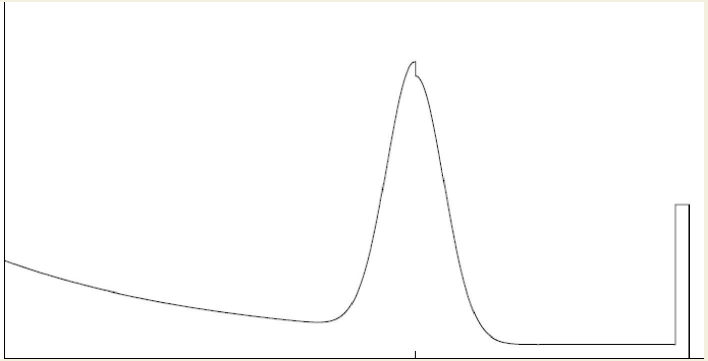
for $m = 1$ *to* M *do*

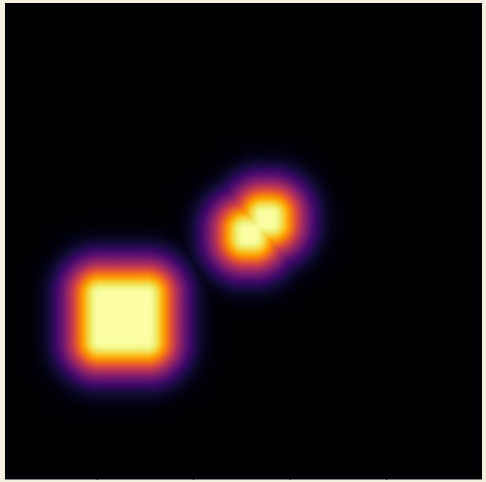
  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

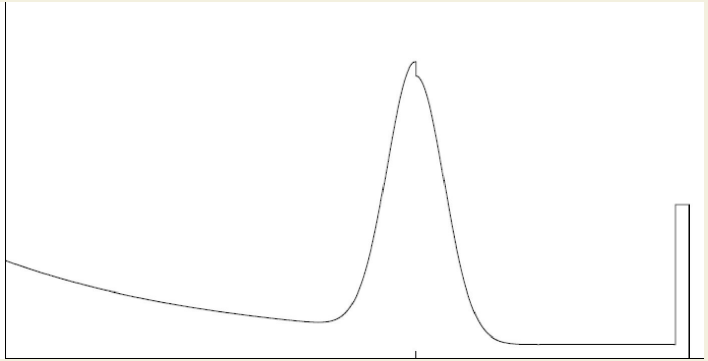
return $\mathcal{X}_t$





$$Bel(x_t) = \eta \boxed{P(z_t \mid x_t)} \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Actualización



Algorithm beam_range_finder_model(z_t, x_t, m):

$q = 1$

for $k = 1$ *to* K *do*

compute z_t^{k*} *for the measurement* z_t^k *using ray casting*

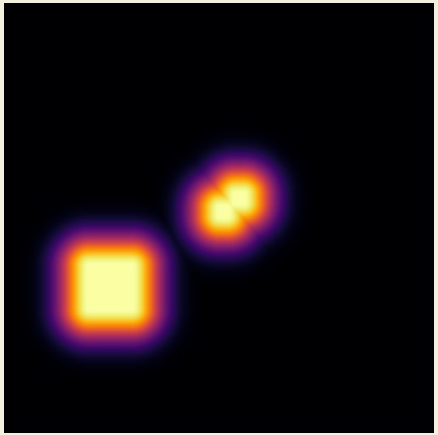
$p = z_{\text{hit}} \cdot p_{\text{hit}}(z_t^k \mid x_t, m) + z_{\text{short}} \cdot p_{\text{short}}(z_t^k \mid x_t, m)$

$+ z_{\text{max}} \cdot p_{\text{max}}(z_t^k \mid x_t, m) + z_{\text{rand}} \cdot p_{\text{rand}}(z_t^k \mid x_t, m)$

$q = q \cdot p$

return q

Filtro de Partículas: Actualización



Algorithm likelihood_field_range_finder_model(z_t, x_t, m)**:**

$q = 1$

for all k *do*

  if $z_t^k \neq z_{\max}$

    $x_{z_t^k} = x + x_{k,\text{sens}} \cos\theta - y_{k,\text{sens}} \sin\theta + z_t^k \cos(\theta + \theta_{k,\text{sens}})$

    $y_{z_t^k} = y + y_{k,\text{sens}} \cos\theta + x_{k,\text{sens}} \sin\theta + z_t^k \sin(\theta + \theta_{k,\text{sens}})$

    $dist = \min_{x',y'} \left\{ \sqrt{(x_{z_t^k} - x')^2 + (y_{z_t^k} - y')^2} \,\middle|\, \langle x', y' \rangle \text{ occupied in } m \right\}$

    $q = q \cdot \left(z_{\text{hit}} \cdot \mathbf{prob}(dist, \sigma_{\text{hit}}) + \frac{z_{\text{random}}}{z_{\max}} \right)$

return q

Filtro de Partículas: Actualización

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

  sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

  $w_t^{[m]} = p(z_t \mid x_t^{[m]})$

  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

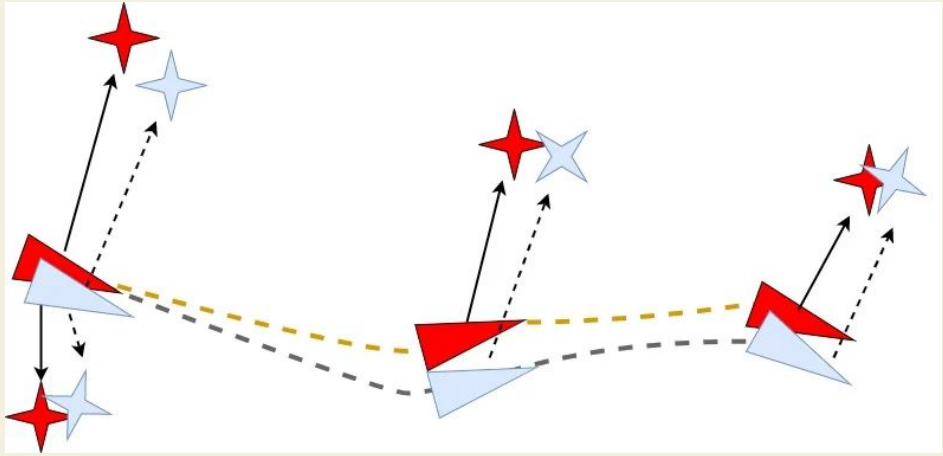
for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

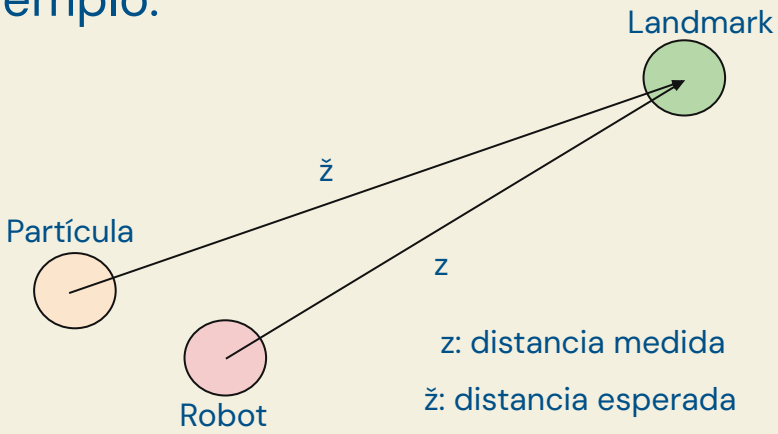
return $\mathcal{X}_t$



$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

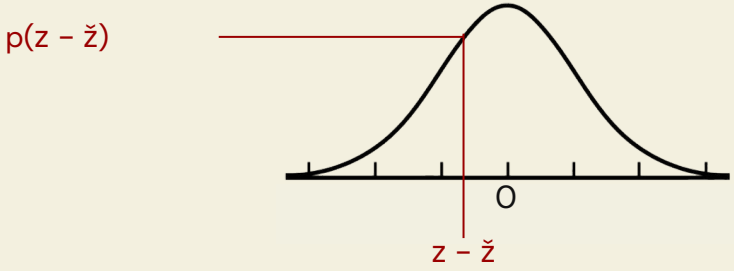
Filtro de Partículas: Actualización

Ejemplo:



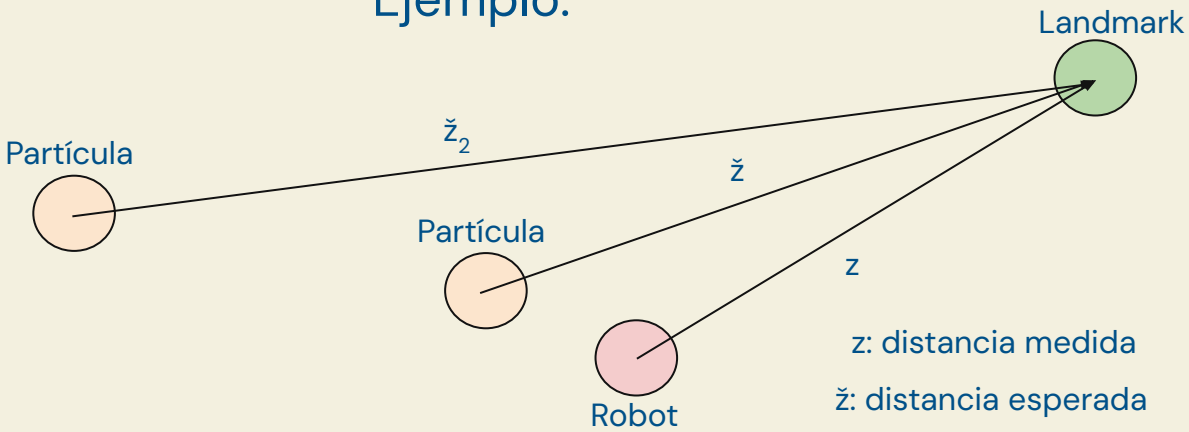
z: distancia medida

ž: distancia esperada



Filtro de Partículas: Actualización

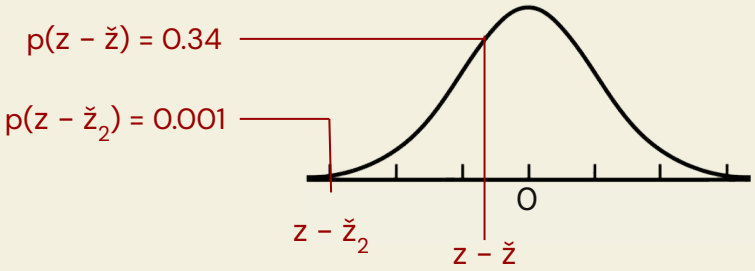
Ejemplo:



z: distancia medida

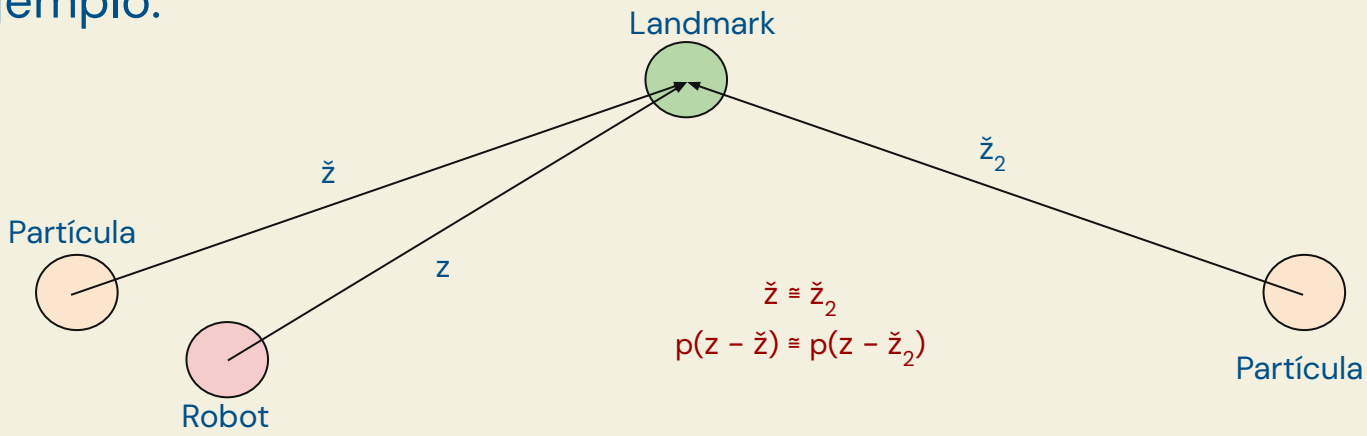
$\check{z}$: distancia esperada

$\check{z}_2$: distancia esperada 2



Filtro de Partículas: Actualización

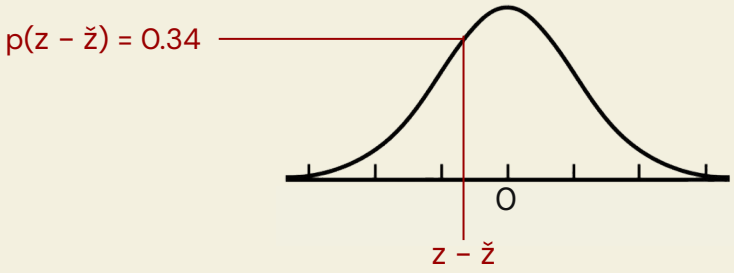
Ejemplo:



z: distancia medida

ž: distancia esperada

$ž_2$: distancia esperada 2



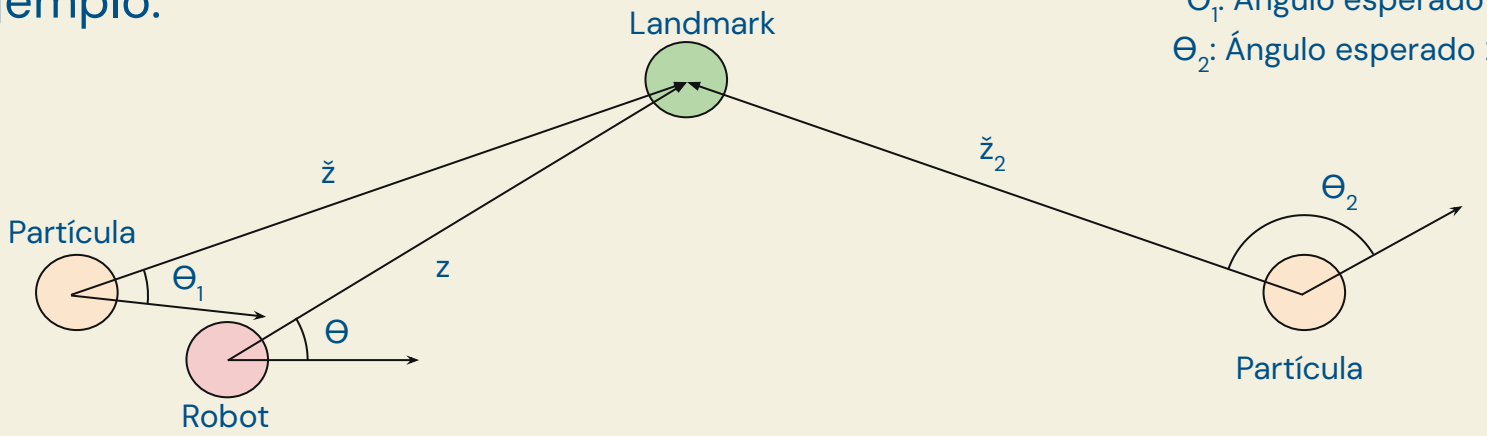
Filtro de Partículas: Actualización

Ejemplo:

Θ: Ángulo medido

Θ_1: Ángulo esperado 1

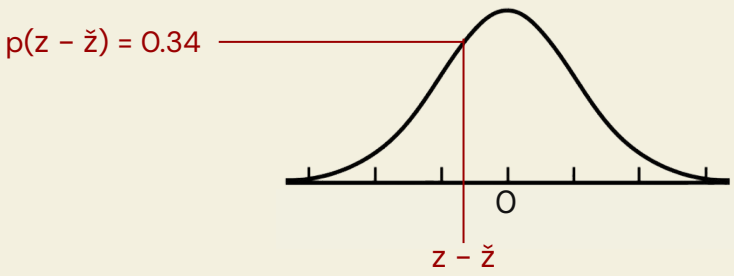
Θ_2: Ángulo esperado 2

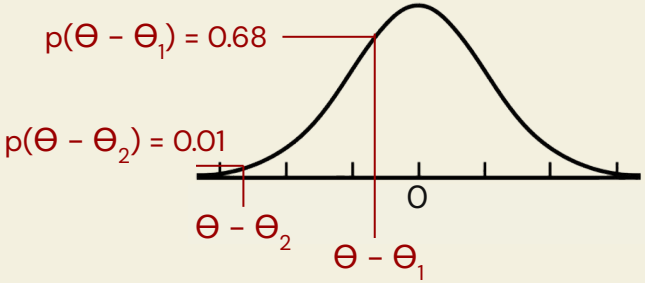


z: distancia medida

ž: distancia esperada

$ž_2$: distancia esperada 2





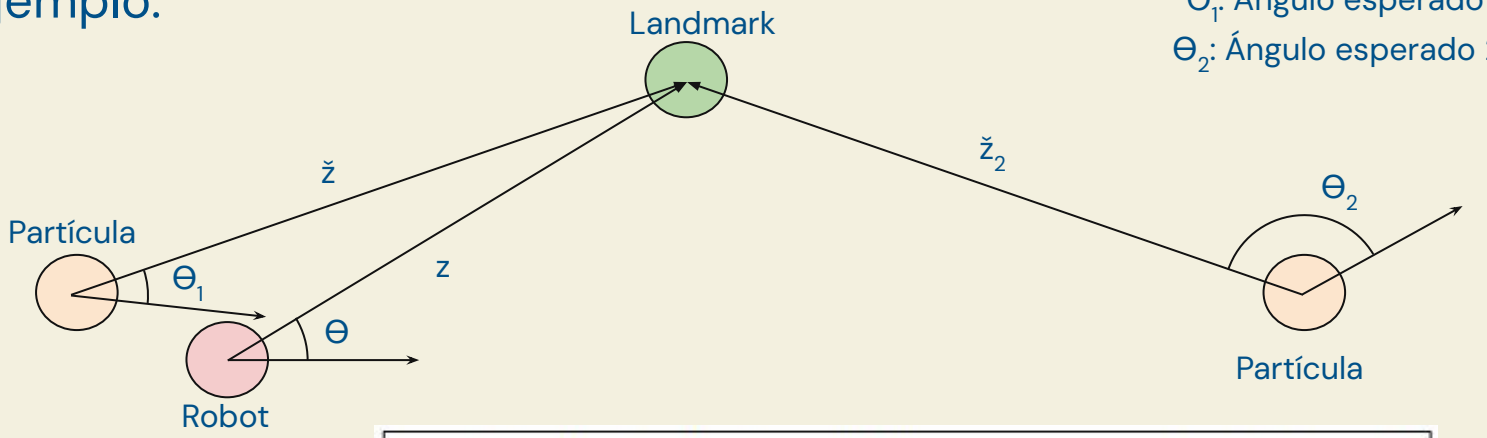
Filtro de Partículas: Actualización

Ejemplo:

Θ: Ángulo medido

Θ_1: Ángulo esperado 1

Θ_2: Ángulo esperado 2



z: distancia medida

ž: distancia esperada

$\check{z}_2$: distancia esperada 2

Algorithm landmark_model_known_correspondence(f_t^i, c_t^i, x_t, m)**:**

$j = c_t^i$

$\hat{r} = \sqrt{(m_{j,x} - x)^2 + (m_{j,y} - y)^2}$

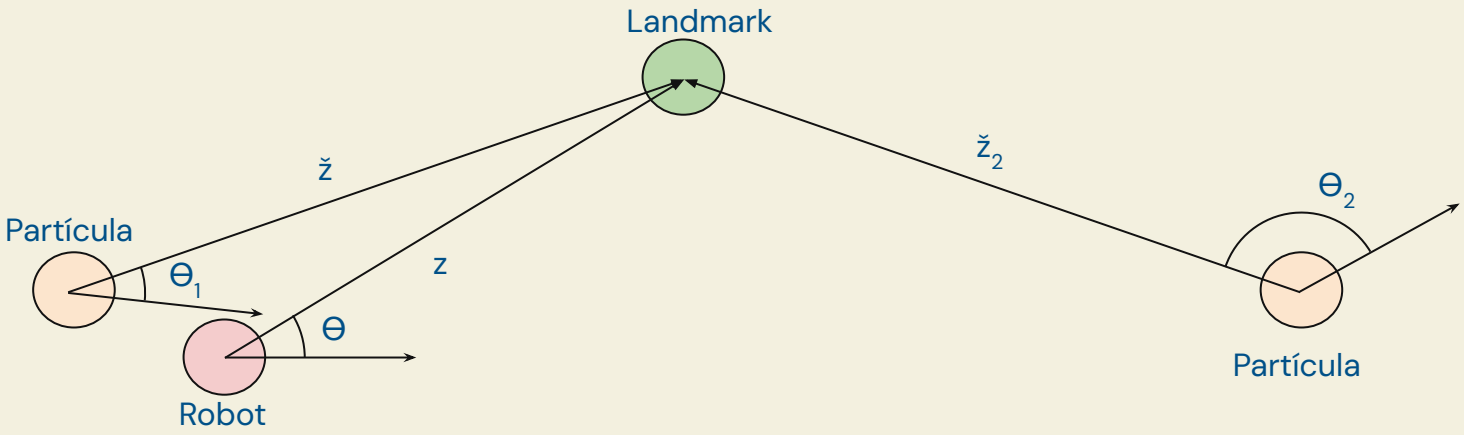
$\hat{\phi} = \text{atan2}(m_{j,y} - y, m_{j,x} - x)$

$q = \textbf{prob}(r_t^i - \hat{r}, \sigma_r) \cdot \textbf{prob}(\phi_t^i - \hat{\phi}, \sigma_\phi)$

return q

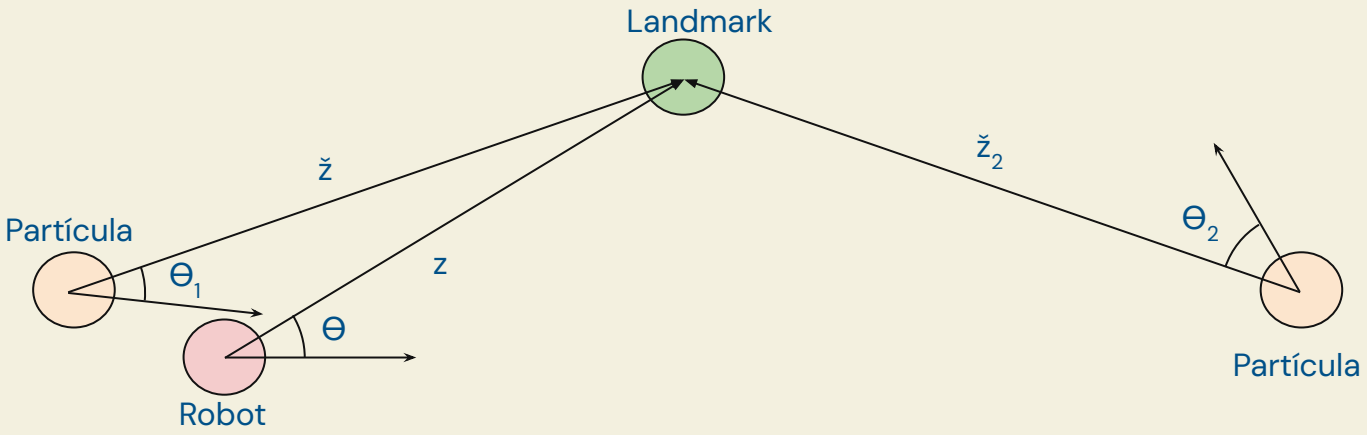
Filtro de Partículas: Actualización

Ejemplo:



Filtro de Partículas: Actualización

Ejemplo:



$\check{z} \cong \check{z}_2$

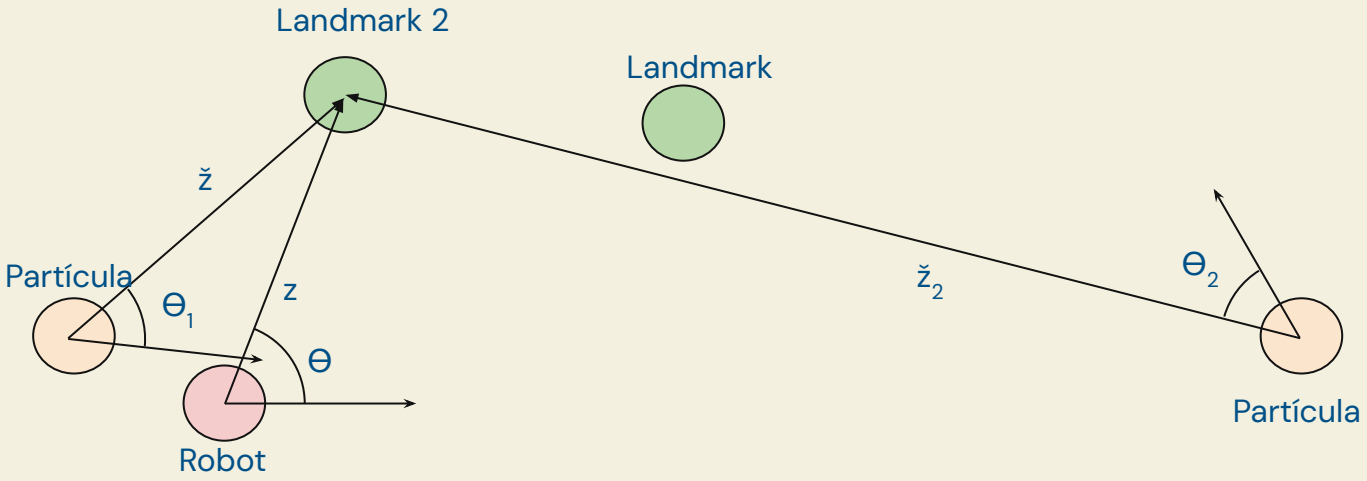
$p(z - \check{z}) \cong p(z - \check{z}_2)$

$\Theta \cong \Theta_1 \cong \Theta_2$

$p(\Theta - \Theta_1) \cong p(\Theta - \Theta_2)$

Filtro de Partículas: Actualización

Ejemplo:

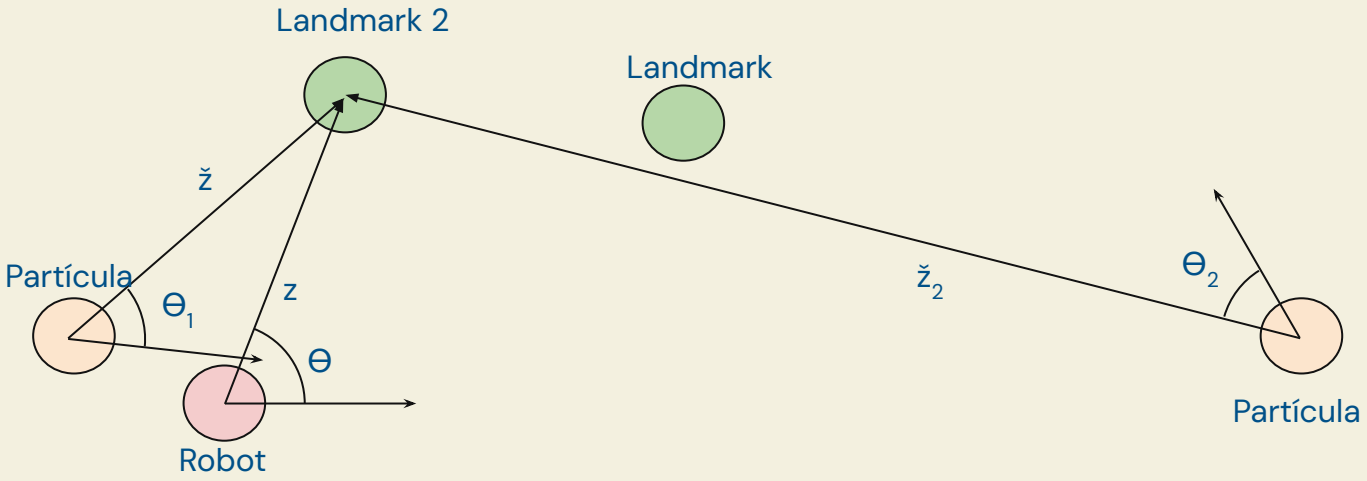


$p(z - \check{z}) >> p(z - \check{z}_2)$

$p(\Theta - \Theta_1) > p(\Theta - \Theta_2)$

Filtro de Partículas: Actualización

Ejemplo:



P(Partícula 1) = P(Landmark 1 | Partícula 1) * P(Landmark 2 | Partícula 1)

P(Partícula 2) = P(Landmark 1 | Partícula 2) * P(Landmark 2 | Partícula 2)

P(Partícula 1) > P(Partícula 2)

Filtro de Partículas: Actualización

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

  sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

  $w_t^{[m]} = p(z_t \mid x_t^{[m]})$

  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

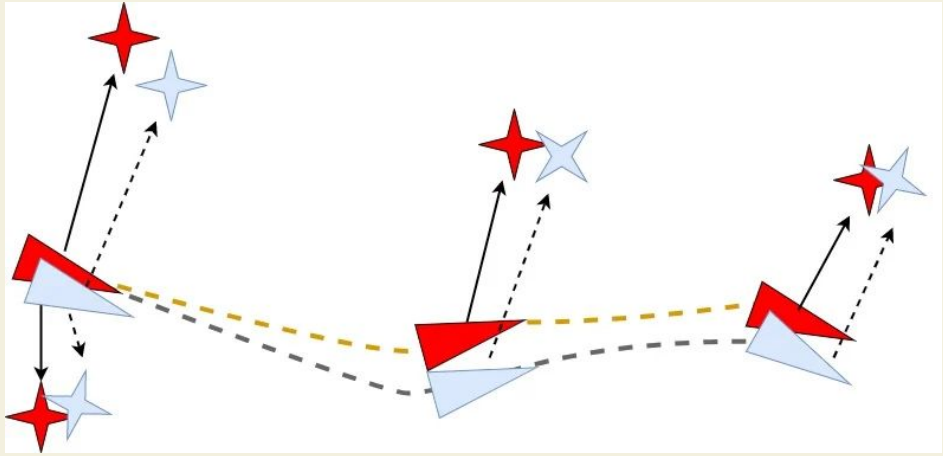
for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$



$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Actualización

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  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

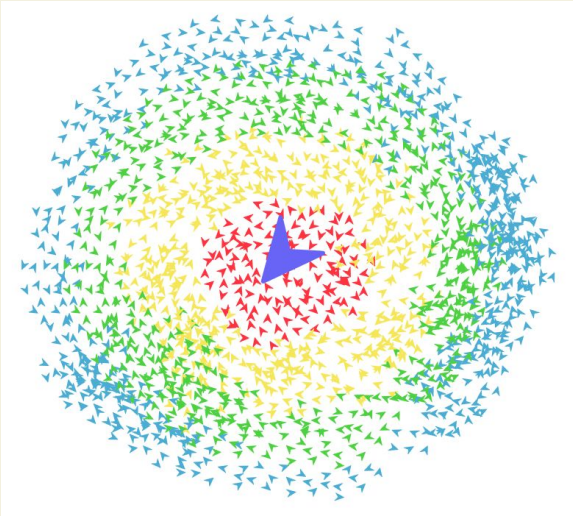
  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

Normalizar



$$Bel(x_t) = \boxed{\eta} P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Actualización

Algorithm Particle_filter$(\mathcal{X}_{t-1}, u_t, z_t)$**:**

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

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  $w_t^{[m]} = p(z_t \mid x_t^{[m]})$

  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

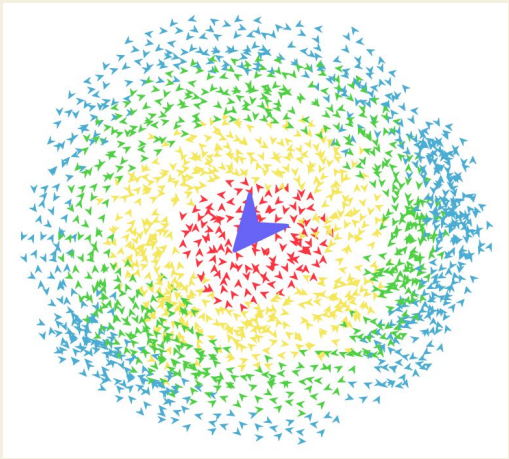
  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

Normalizar

SUM



$\neq 1$

$$Bel(x_t) = \boxed{\eta} P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Actualización

Algorithm Particle_filter$(\mathcal{X}_{t-1}, u_t, z_t)$**:**

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

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  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

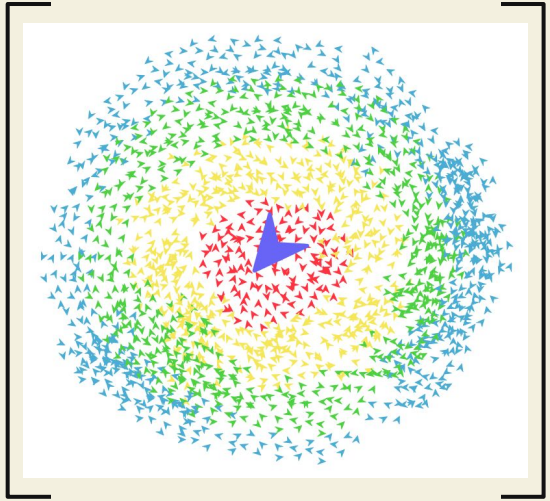
return $\mathcal{X}_t$

$$Bel(x_t) = \boxed{\eta} P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Normalizar

$$\frac{\text{Peso de particula}}{\text{Suma de todos los pesos}}$$

SUM



= 1

Filtro de Partículas: Remuestreo

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

  sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

  $w_t^{[m]} = p(z_t \mid x_t^{[m]})$

  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

Resampleo: elegir modelo

Ejemplo:

- Saco todos los pesos menores a 5%
  - Criterio de lenta convergencia → muy exigente
- Stochastic Universal Sampling (SUS)

$$Bel(x_t) = \boxed{\eta} P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Remuestreo

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

$w_t^{[m]} = p(z_t \mid x_t^{[m]})$

$\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

draw i *with probability* $\propto w_t^{[i]}$

add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

Stochastic Universal Sampling: algorítmo genético

1. Algorithm **systematic_resampling**(S,n):

2. $S' = \varnothing, c_1 = w^1$
3. **For** $i = 2 \ldots n$ — *Generate cdf*
4. $c_i = c_{i-1} + w^i$
5. $u_1 \sim U[0, n^{-1}], i = 1$ — *Initialize threshold*

6. **For** $j = 1 \ldots n$ — *Draw samples …*
7. **While** ($u_j > c_i$) — *Skip until next threshold reached*
8. $i = i + 1$
9. $S' = S' \cup \{< x^i, n^{-1} >\}$ — *Insert*
10. $u_{j+1} = u_j + n^{-1}$ — *Increment threshold*

11. **Return** S'

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Remuestreo

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

  sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

  $w_t^{[m]} = p(z_t \mid x_t^{[m]})$

  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

Stochastic Universal Sampling: algorítmo genético

1. Algorithm **systematic_resampling**(S,n):

2. $S' = \emptyset, c_1 = w^1$
3. **For** $i = 2 \ldots n$ — *Generate cdf*
4. $c_i = c_{i-1} + w^i$ → Cumsum, e.g.: cs_weights[0.1 0.3 0.6 0.9 1]
5. $u_1 \sim U[0, n^{-1}], i = 1$ — *Initialize threshold*

6. **For** $j = 1 \ldots n$ — *Draw samples …*
7. **While** ($u_j > c_i$) — *Skip until next threshold reached*
8. $i = i + 1$
9. $S' = S' \cup \{< x^i, n^{-1} >\}$ — *Insert*
10. $u_{j+1} = u_j + n^{-1}$ — *Increment threshold*

11. **Return** S'

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Remuestreo

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

  sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

  $w_t^{[m]} = p(z_t \mid x_t^{[m]})$

  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

Stochastic Universal Sampling: algorítmo genético

1. Algorithm **systematic_resampling**(S,n):

2. $S' = \emptyset, c_1 = w^1$
3. **For** $i = 2 \ldots n$ — ***Generate cdf***
4. $c_i = c_{i-1} + w^i$
5. $u_1 \sim U[0, n^{-1}], i = 1$ — ***Initialize threshold***

Umbral (threshold) proporcional a una uniforme entre 0 y la inversa del número de partículas (1/n):

- seed = U[0, 1/n]
- u_j = seed + j * 1/n

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Remuestreo

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endfor

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add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

Stochastic Universal Sampling: algorítmo genético

1. Algorithm **systematic_resampling**(*S*,*n*):

Lógica: si u_j < c_i la partícula sobrevive y entra en la lista, de lo contrario pasamos a la siguiente (i++)

6. **For** $j = 1 \ldots n$ — *Draw samples …*
7. **While** ($u_j > c_i$) — *Skip until next threshold reached*
8. $i = i + 1$
9. $S' = S' \cup \{< x^i, n^{-1} >\}$ — *Insert*
10. $u_{j+1} = u_j + n^{-1}$ — *Increment threshold*
11. **Return** S'

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Filtro de Partículas: Remuestreo

Algorithm Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

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  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Stochastic Universal Sampling: algorítmo genético

1. Algorithm **systematic_resampling**(*S*,*n*):

Ejemplo:

Cs_weights = [0.1 0.3 0.35 0.36 0.37 0.6 0.8 1]

X_hat = [p1 p2 p3 p4 p5 p6 p7 p8]

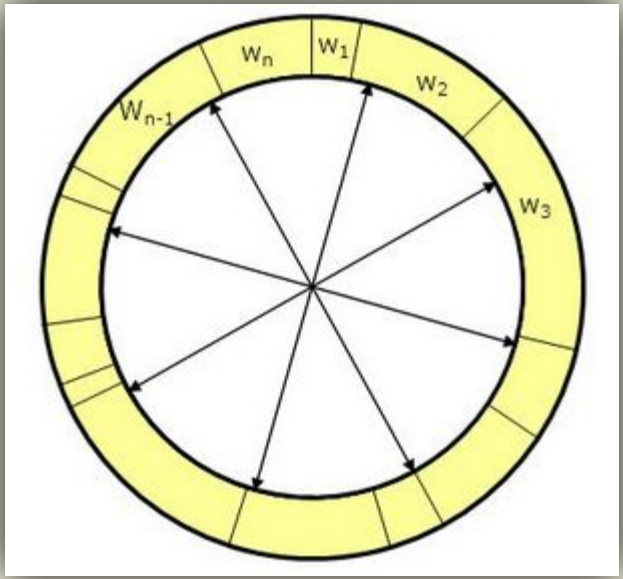
seed = 0.125

u_j = U(0, 0.125)+ j * 0.125 = [] = 0.05 + j * 0.12

u_j = [0.175 0.3 0.425 0.55 0.675 0.8 0.925 1.05]

X = [p2 p2 p6 p6 p7 p8 p8 p8]

Stochastic Universal Sampling



Stochastic Universal Sampling

p1 : 0.2

p2 : 0.1

p3 : 0.15

p4 : 0.05

p5 : 0.05

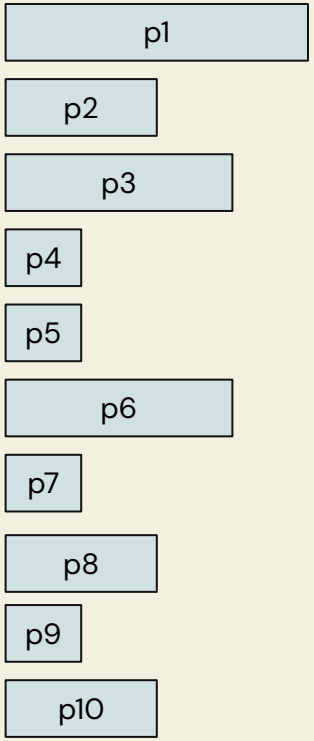
p6 : 0.15

p7 : 0.05

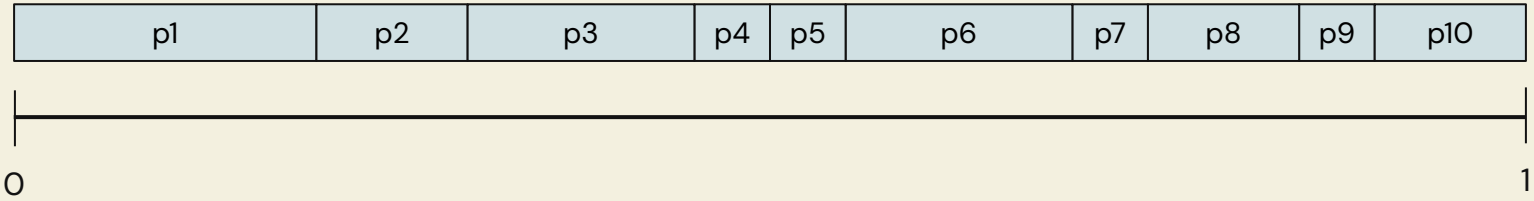
p8 : 0.1

p9 : 0.05

p10 : 0.1



Stochastic Universal Sampling

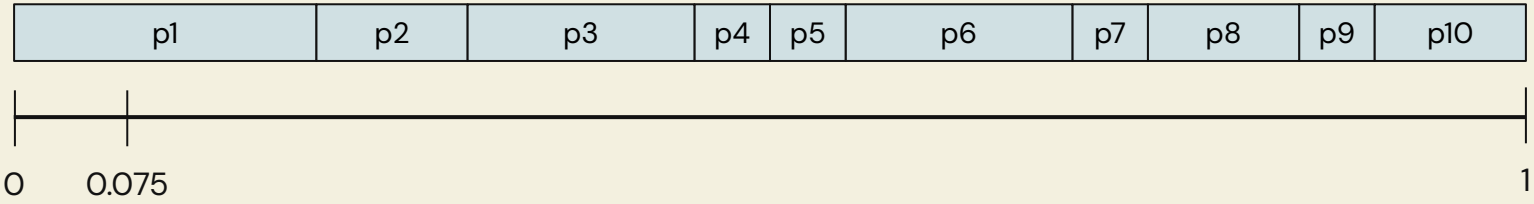


n = número de particulas a samplear

1/n = 0.1

Eligo punto inicial → [0, 1/n)

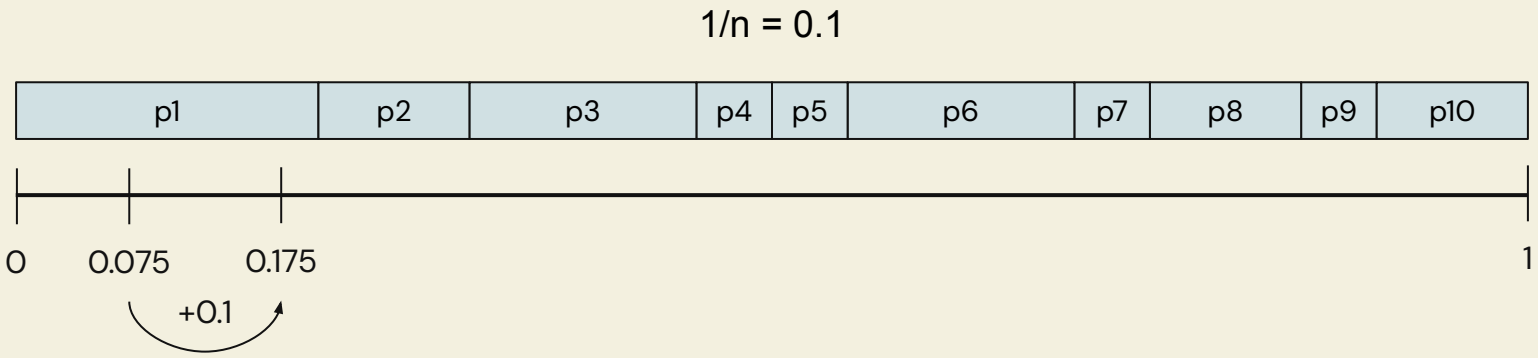
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1,

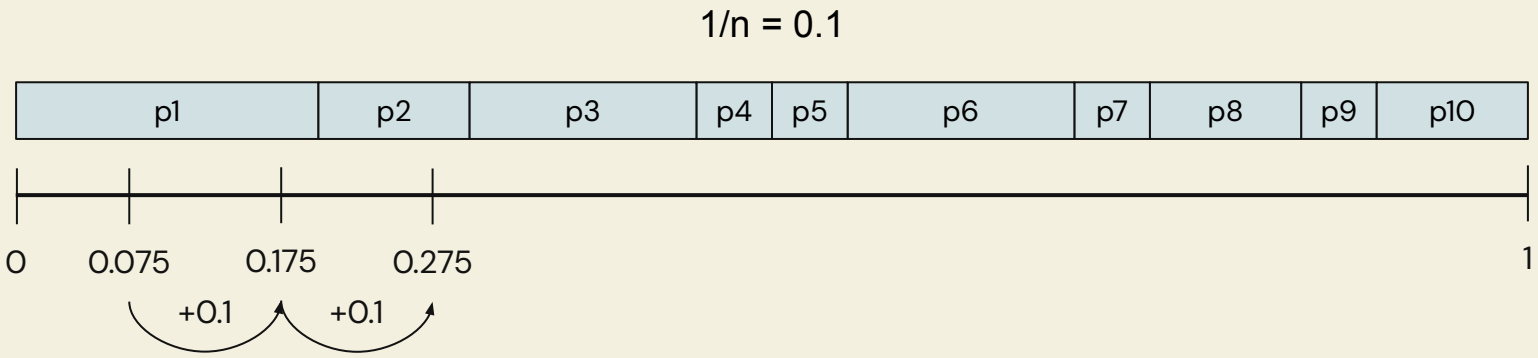
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1,

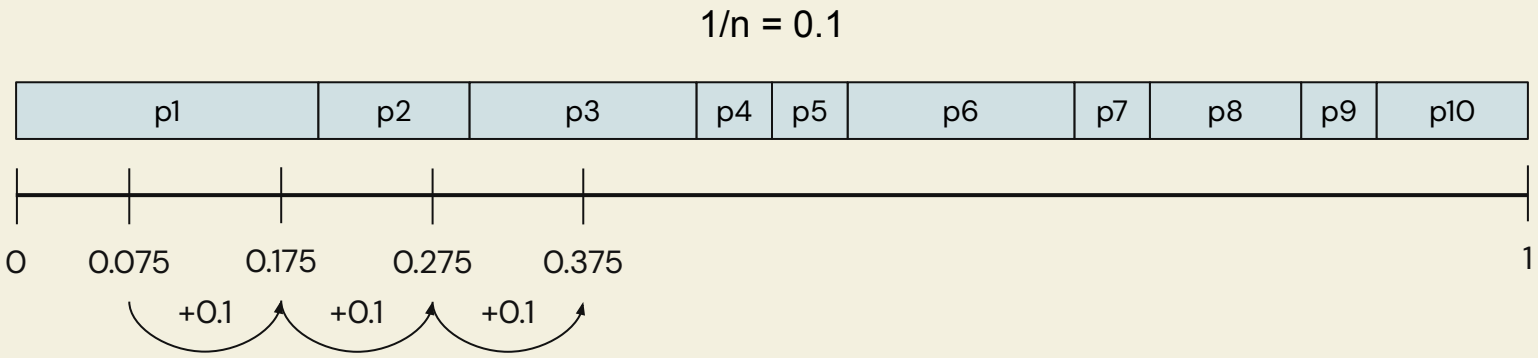
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2,

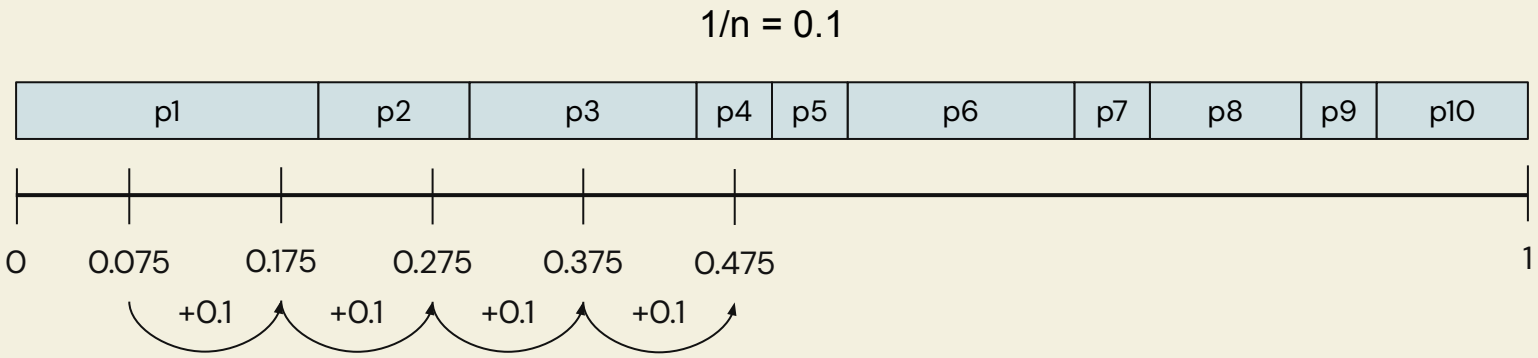
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2, p3

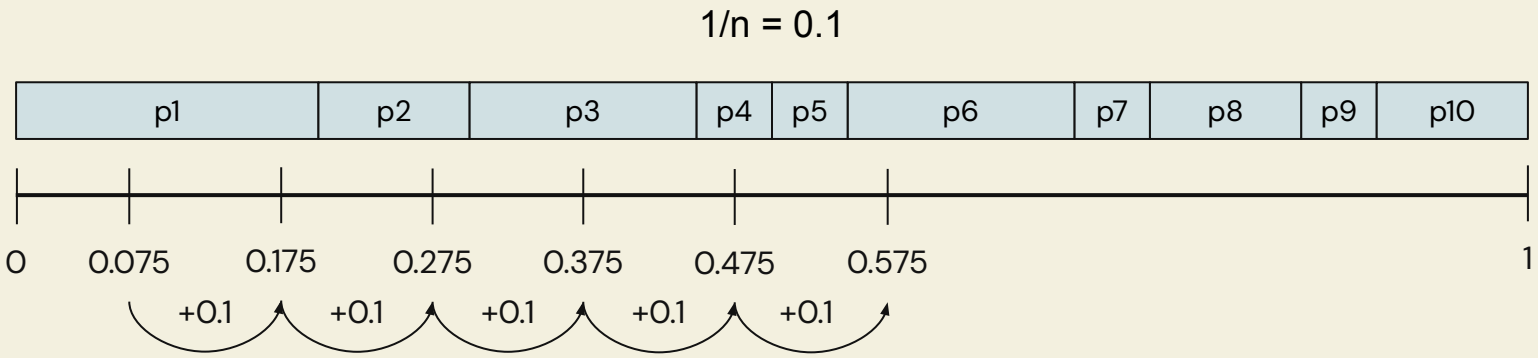
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2, p3, p4

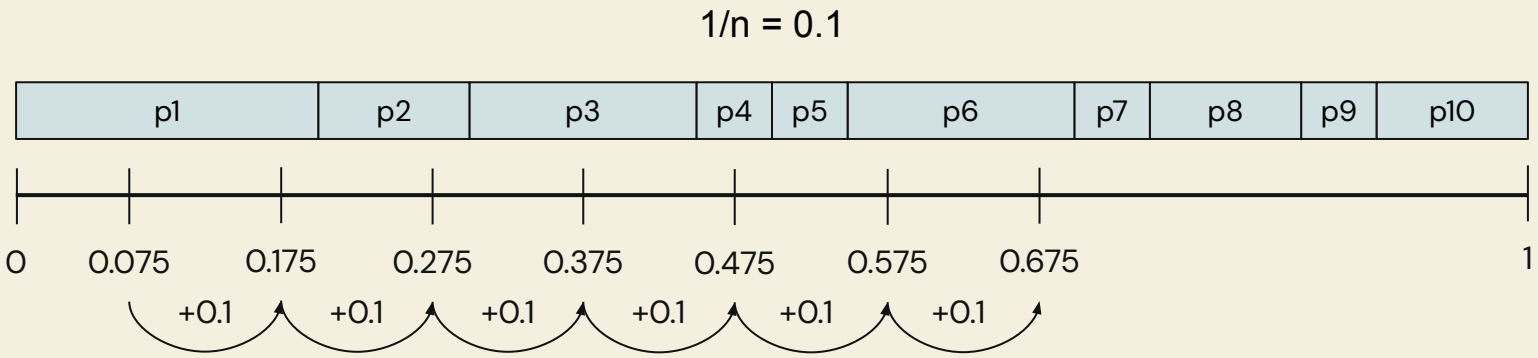
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2, p3, p4, p6

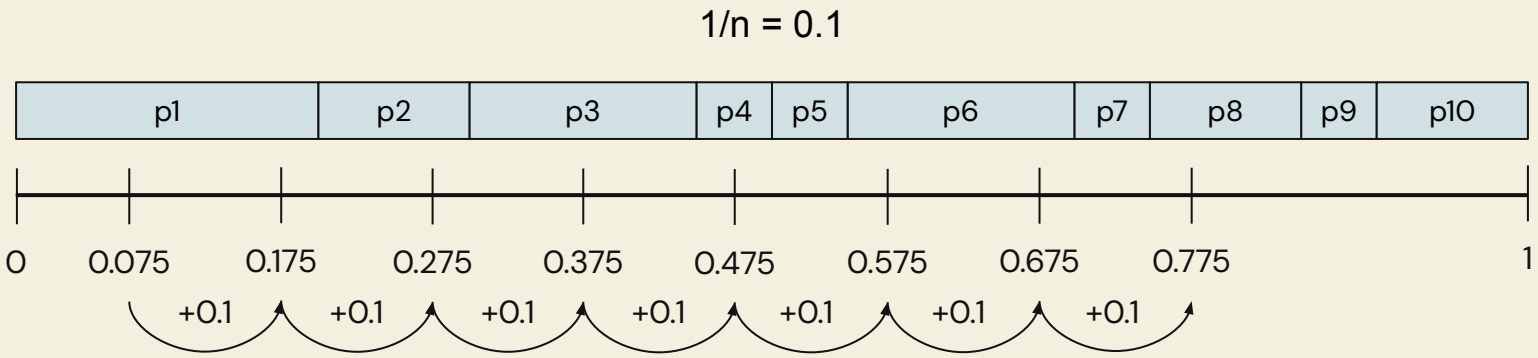
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2, p3, p4, p6, p6

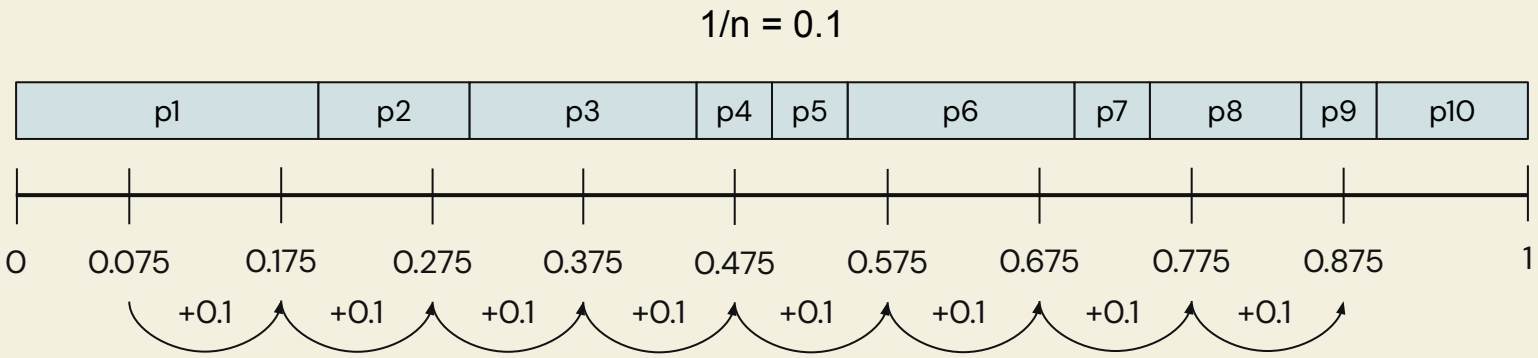
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2, p3, p4, p6, p6, p8

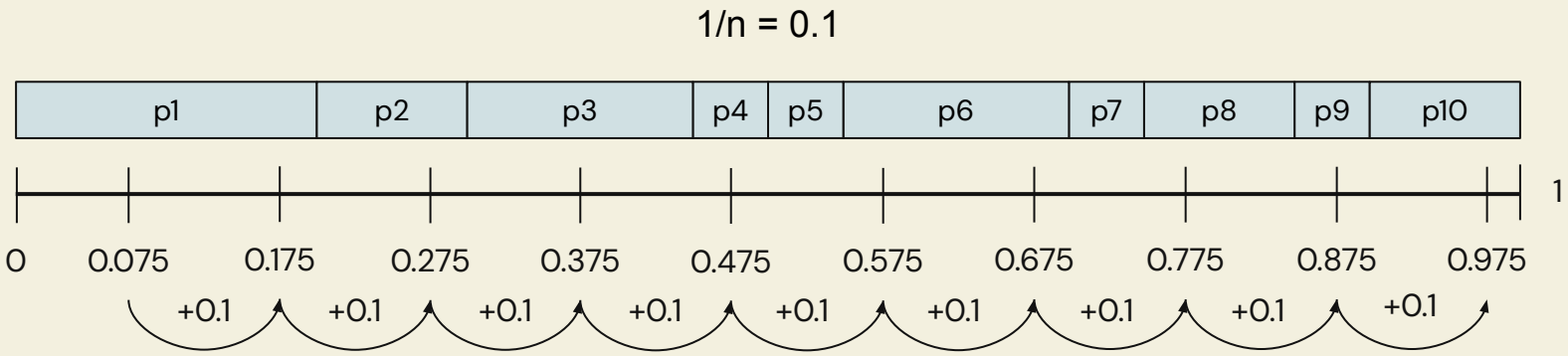
Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2, p3, p4, p6, p6, p8, p9

Stochastic Universal Sampling



Nuevas partículas (copias de):

p1, p1, p2, p3, p4, p6, p6, p8, p9, p10

Filtro de Partículas

Algorithm Particle_filter$(\mathcal{X}_{t-1}, u_t, z_t)$**:**

$\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

for $m = 1$ *to* M *do*

  sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$

  $w_t^{[m]} = p(z_t \mid x_t^{[m]})$

  $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$

endfor

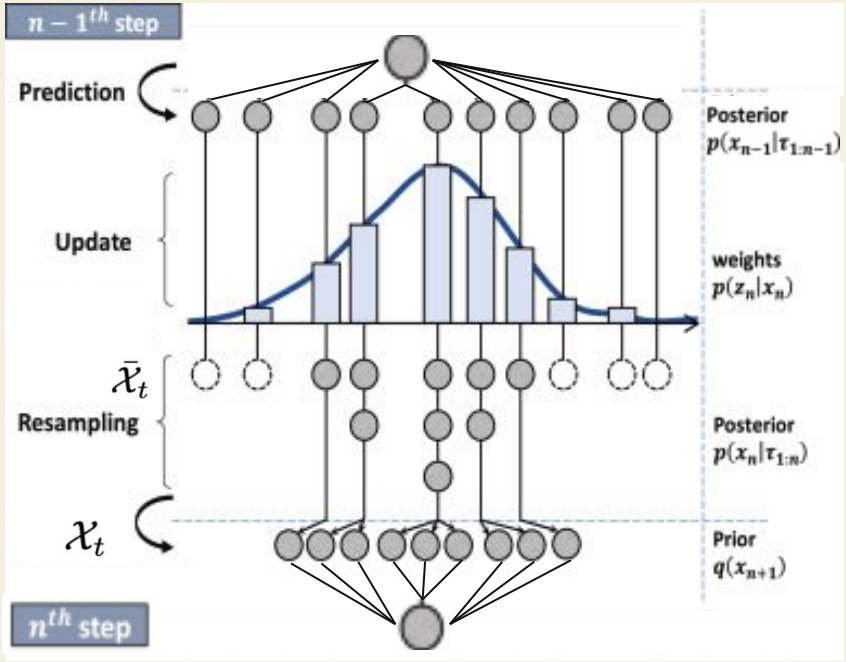
for $m = 1$ *to* M *do*

  draw i *with probability* $\propto w_t^{[i]}$

  add $x_t^{[i]}$ *to* $\mathcal{X}_t$

endfor

return $\mathcal{X}_t$



$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$